

1	1	/	P?S/8 SYSTEM HANDLER TD8E/MR8E-C	1
2	2			2
3	3	/	COPYRIGHT (C) 1975-1989, CHARLES LASNER ASSOCIATES; 1990-2016, CLA SYSTEMS.	3
4	4			4
5	5	/	LAST EDIT: 05-MAR-2016 12:00:00 CJL	5
6	6			6
7	7	/	MUST BE ASSEMBLED WITH '/J' [OS/8 PAL8 '/F'] OPTION SWITCH SET.	7
8	8			8
9	9	/	P?S/8 SYSTEM DEVICE HANDLER FOR PRIMARY TD8E [DEVICE CODE 77] WITH TU56H [ONE	9
10	10	/	DRIVE ONLY] OR TU56 [TWO DRIVES].	10
11	11			11
12	12	/	AUTOMATICALLY REMEMBERS PROPER INITIAL SEARCH DIRECTION WHEN THE LOGICAL	12
13	13	/	CONSOLE OVERLAY IS IN EFFECT; OTHERWISE ALWAYS STARTS SEARCHES IN REVERSE	13
14	14	/	SIMILAR TO TC12. [NOTE: THIS APPLIES TO CALLS FOR EITHER LOGICAL DRIVE UNIT.]	14
15	15			15
16	16	/	IGNORES THE TWO HIGH-ORDER BITS OF THE LOGICAL UNIT FOR ALL CALLS. THUS, THE	16
17	17	/	CALL CANNOT DISTINGUISH BETWEEN DRIVES 0, 2, 4, 6 OR 1, 3, 5, 7; FUTURE	17
18	18	/	RELEASES WILL CAUSE A SELECT ERROR ON LOGICAL UNITS 2 THROUGH 7 WHEN THE	18
19	19	/	LOGICAL CONSOLE OVERLAY IS IN EFFECT. [NOTE: THE MR8E-C ROM ONLY SUPPORTS THE	19
20	20	/	DEVICE 77 TD8E CONTROLLER; NON-SYSTEM DEVICE HANDLERS CAPABLE OF SUPPORTING 4	20
21	21	/	TD8E CONTROLLERS AND UP TO EIGHT DRIVES WILL EVENTUALLY BE IMPLEMENTED.]	21
22	22			22
23	23	/	CAN ONLY BE BOOTSTRAPPED TO LOGICAL DRIVE UNIT 0. THIS IS A RESTRICTION OF	23
24	24	/	THE MR8E-C ROM. ALTERNATE BOOTSTRAPS MAY BE DEVELOPED IN FUTURE RELEASES TO	24
25	25	/	OVERCOME THIS [WITH DIFFICULTY].	25
26	26			26
27	27	/	THE BASIC DEVICE HANDLER HALTS/RETRIES ON ALL ERRORS DETECTED BY THE ROM. DUE	27
28	28	/	TO LACK OF SPACE IN FIELD 0, THE ROM-BASED RETRY COUNT LOGIC IS NOT SUPPORTED.	28
29	29	/	HOWEVER, IN KEEPING WITH THE P?S/8 PHILOSOPHY OF NOT OBFUSCATING HARDWARE	29
30	30	/	PROBLEMS, THIS DESIGN DECISION WOULD APPLY EVEN IF THE CODE SPACE WERE	30
31	31	/	AVAILABLE. IT IS NECESSARY TO MAKE THE USER AWARE OF DEGRADING HARDWARE	31
32	32	/	SITUATIONS; THIS CAN ONLY BE REALIZED BY ALERTING THE USER WITH APPROPRIATE	32
33	33	/	ERROR HANDLING. WHEN THE LOGICAL CONSOLE OVERLAY IS IN EFFECT, THE NATURE OF	33
34	34	/	THE ERROR WILL BE EVEN MORE OBVIOUS. HOWEVER, PROPERLY FUNCTIONING EQUIPMENT	34
35	35	/	AND DECTAPE MEDIA DOES NOT ROUTINELY CAUSE ERRORS; THEY SHOULD BE TREATED AS A	35
36	36	/	SIGN OF IMPENDING FAILURE OF THE DRIVE GUIDES AND/OR THE DECTAPE MEDIA. IF	36
37	37	/	TAKEN CARE OF EARLY, ALL DATA ON THE DECTAPE SHOULD BE RECOVERABLE BEFORE THE	37
38	38	/	PROBLEM WORSENS. [NOTE: THE RETRY COUNT IS USED THROUGHOUT OS/8 DECTAPE	38
39	39	/	HANDLING AND IS THE ROOT CAUSE OF THIS PROBLEM; BY PROPER DESIGN, THIS	39
40	40	/	"FEATURE" WITHIN THE MR8E-C IS AVOIDED.]	40
41	41			41
42	42	/	INTERRUPT STATUS IS RESTORED AFTER A CALL TO THE SYSTEM HANDLER; INTERNALLY,	42
43	43	/	THE HANDLER OPERATES WITH INTERRUPTS OFF TO PREVENT TIMING PROBLEMS. WHILE	43
44	44	/	THERE IS NO RULES REGARDING THIS THROUGHOUT P?S/8, GIVEN THAT THE TD8E IS TO	44
45	45	/	BE USED ONLY ON OMNIBUS MACHINES ALLOWS THIS FEATURE TO BE ADDED. [NOTE: DUE	45
46	46	/	TO SPEED CONSIDERATIONS, IT IS RECOMMENDED THE TD8E ONLY BE USED WHERE THE	46
47	47	/	KK8E CPU IS AVAILABLE [PDP-8/E, PDP-8/F, PDP-8/M AND PDP-8/A-600 SERIES].	47
48	48	/	THIS IS ESPECIALLY PROBLEMATIC WHEN THE MR8E-C IS IN USE DUE TO THE OVERHEAD	48
49	49	/	CAUSED BY INTER-FIELD SUBROUTINE CALLS.	49
50	50			50
51	51	/	ADDITIONAL NOTEWORTHY FEATURES WILL BE DESCRIBED HERE AS THE BINARY IMAGE CODE	51
52	52	/	IS DISSASSEMBLED.	52
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1	54	/	NOTE: CERTAIN FEATURES OF THE HANDLER ARE BASED ON THE RECOLLECTION OF	1
2	55	/	CHARLES LASNER, P?S NY, WHO HASTILY WROTE THE ORIGINAL VERSION 40 YEARS AGO,	2
3	56	/	WHEN BRIEFLY AN APPROPRIATE PDP-8/E WITH TD8E AND MR8E-C CONFIGURATION WAS	3
4	57	/	AVAILABLE DURING A TRANSIENT PROGRAMMING ASSIGNMENT. DUE TO CIRCUMSTANCES,	4
5	58	/	THIS SYSTEM CONFIGURATION WAS NEVER UPDATED; THE ORIGINAL SOURCE CODE IS NOT	5
6	59	/	AVAILABLE AND PRESUMED PERMANENTLY LOST. IT IS KNOWN THAT THE SOURCE CODE OF	6
7	60	/	THE MR8E-C ROM WAS INCLUDED WITHIN THE SOURCE FILE OF THE HANDLER TO ALLOW	7
8	61	/	OCCASIONAL DEVELOPMENT CONSISTENT WITH THE OVERALL EVOLVEMENT OF P?S/8.	8
9	62	/	HOWEVER, NO SIGNIFICANT CHANGES TO P?S/8 OF ANY RELEVANCE TO THE TD8E WERE	9
10	63	/	MADE AFTER THE LAST KNOWN CONTACT WITH AN ALTERNATE PDP-8/E SYSTEM IN 1976	10
11	64	/	[AND PERHAPS THE EARLY PART OF 1977. THE LAST KNOWN SYSTEM CONSISTED OF A	11
12	65	/	TU56H, TD8E AND 32K OF CORE MEMORY [AND UNRELATED PERIPHERALS]. THE MR8E-C	12
13	66	/	ROM IMAGE WAS LOADED BEFORE BOOTING THE SYSTEM TO ALLOW CREATING AN IMAGE OF A	13
14	67	/	FUNCTIONAL SYSTEM FOR TESTING.	14
15	68			15
16	69	/	DUE TO THE CIRCUMSTANCES OF P?S/8 DEVELOPMENT, ALL TD8E-SPECIFIC CODE WAS	16
17	70	/	STRIPPED OUT OF THE TYPICAL SET OF DEVELOPMENTAL SOURCE FILES DUE TO FILE SIZE	17
18	71	/	LIMITATIONS. DUE TO THE APPLICABLE LOGISTICS, ALL DEVELOPMENT FILES WERE	18
19	72	/	SPLIT INTO TWO SECTIONS EACH CAPABLE OF BEING STORED ON HALF OF AN RK8E/RK05	19
20	73	/	PACK. IT WAS IMPRACTICAL TO PHYSICALLY TRANSPORT THE MEDIA; DECISIONS WERE	20
21	74	/	MADE TO SPLIT OFF THE TD8E-RELATED FILES [AS WELL AS CERTAIN OTHER FILES	21
22	75	/	RELEVANT TO OTHER CONFIGURATIONS]. IT IS HOPED THE SOURCE FILES FOR THESE	22
23	76	/	OTHER CONFIGURATIONS CAN BE RECONSTRUCTED FROM SURVIVING BINARY MEDIA; IF THIS	23
24	77	/	CANNOT BE ACCOMPLISHED, THE CONFIGURATION WILL REQUIRE COMPLETE SOURCE	24
25	78	/	RECREATION AND ACCESS TO THE APPLICABLE HARDWARE AT THE MINIMUM. NOTE:	25
26	79	/	CERTAIN ENHANCEMENTS TO THE SOURCE CODE SPECIFIC TO THESE SYSTEMS WILL ALSO	26
27	80	/	REQUIRE ADDITIONAL CONTACT WITH THE RELEVANT HARDWARE; P?S/8 DEVELOPMENT	27
28	81	/	IS NOW ACTIVE AGAIN AFTER A SHORT RECESS OF ABOUT 22 YEARS. SOME OF THESE	28
29	82	/	ISSUES WILL BE RESOLVED AS TIME PERMITS AS WELL AS ACCESS TO THE RELEVANT	29
30	83	/	HARDWARE. THE TD8E ISSUES WILL BE RESOLVED USING THE SIMH PDP-8 SIMULATOR	30
31	84	/	FOLLOWED BY PERIODIC TESTING ON TD8E-BASED PDP-8/E HARDWARE.	31
32	85			32
33	86	/	UNTIL ALL TD8E-SPECIFIC CODE CAN BE DISASSEMBLED FROM THE RECENTLY RECOVERED	33
34	87	/	VERSION 8T DECTAPE, IT IS NOT CLEAR IF THIS TAPE REPRESENTS THE FINAL POINT IN	34
35	88	/	THE DEVELOPMENT OF THIS SYSTEM. CERTAIN COMPONENTS NOT YET CONFIRMED SHOULD	35
36	89	/	PROVIDE PLAUSIBLE ANSWERS. [IT IS KNOWN THERE WAS A SPECIAL INTERACTION	36
37	90	/	BETWEEN THIS HARDWARE CONFIGURATION AND THE LOGICAL CONSOLE OVERLAY WHEN	37
38	91	/	CONFIGURED EXPRESSLY FOR THE VT8E.]	38
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1	93	/	PROGRAM EDIT HISTORY.	1
2	94			2
3	95	/	01-APR-1975 - ORIGINAL RELEASE OF THE BINARY CODE [ESTIMATED]; THIS	3
4	96	/	IS LIKELY THE ONLY SURVIVING COPY OF THE P?S/8	4
5	97	/	OPERATING SYSTEM FOR THE TD8E/MR8E-C. MOST OF THE	5
6	98	/	SOURCE CODE WILL BE OBTAINED BY DISASSEMBLY OF THIS	6
7	99	/	RELEASE RECOVERED ON A SURVIVING DECTAPE.	7
8	100			8
9	101	/	NOTE: IT IS SUSPECTED THIS IS THE ONLY KNOWN SURVIVING	9
10	102	/	COPY OF THE CONFIGURATION FOR THE TD8E WITH MR8E-C	10
11	103	/	ROM. IT IS LIKELY IT WAS NOT PROPERLY TESTED. AS OF	11
12	104	/	THIS WRITING, IT IS NOW KNOWN THE CONFIGURATION IS	12
13	105	/	FATALLY FLAWED, BUT TENTATIVELY FUNCTIONAL BY A	13
14	106	/	CONTRIVED KLUDGE THAT WAS APPARENTLY APPLIED AFTER	14
15	107	/	USING ANY P?S/8 OPERATION THAT MODIFIED BLOCK 0000.	15
16	108	/	LATER RELEASES WILL OVERCOME THIS BY MAKING FURTHER	16
17	109	/	CHANGES IN THE DESIGN WHICH WILL CORRECT THE PROBLEM.	17
18	110	/	THE ONLY KNOWN PROBLEM THAT CANNOT BE OVERCOME IS THAT	18
19	111	/	THE DESIGN CHANGE WILL ELIMINATE THE FATAL PROBLEM,	19
20	112	/	BUT THE SYSTEM BOOTUP PROCEDURE WILL REQUIRE AN	20
21	113	/	ADDITIONAL TAPE TURN-AROUND. [USERS OF THE TC01/TC08	21
22	114	/	RELEASE OF P?S/8 MAY NOTICE THIS TINY LOSS OF	22
23	115	/	PERFORMANCE; BY CONTRAST, OS/8 BOOTUP TAKES MUCH MORE	23
24	116	/	TIME THAN EITHER RELEASE.]	24
25	117			25
26	118	/	18-DEC-2015 - IMAGE OF THE 1975 RELEASE AVAILABLE FOR USE WITH THE	26
27	119	/	SIMH PDP-8 SIMULATOR. EXAMINATION OF THE TAPE	27
28	120	/	CONFIRMS IT IS A SYSTEM CREATED SOMETIME IN 1974/1975.	28
29	121	/	HOWEVER, THE TAPE CANNOT BE BOOTED BECAUSE THE IMAGE	29
30	122	/	OF THE MR8E-C ROM IS NOT AVAILABLE TO THE SIMH PDP-8	30
31	123	/	SIMULATOR.	31
32	124			32
33	125	/	26-JAN-2016 - OPERATIONAL PRE-FINAL RELEASE OF THE SOURCE CODE OF	33
34	126	/	THE MR8E-C ROM IS COMPLETED. USING THE MR8E-C BINARY	34
35	127	/	IMAGE, THE ORIGINAL RELEASE IS BOOTED ON THE SIMH	35
36	128	/	PDP-8 SIMULATOR; ALTHOUGH THIS IS A VERY OLD RELEASE	36
37	129	/	OF P?S/8 [APPARENTLY VERSION 8T BASED ON ASSOCIATED	37
38	130	/	DECTAPE PACKAGING], MOST OF THE BASIC ELEMENTS OF	38
39	131	/	DEVICE-SPECIFIC FUNCTIONALITY ARE CONFIRMED WORKING ON	39
40	132	/	THE SIMH PDP-8 SIMULATOR [OTHER THAN OPERATIONS THAT	40
41	133	/	MODIFY THE CONTENTS OF BLOCK 0000 SUCH AS THE "DATE"	41
42	134	/	COMMAND, WHICH WRITES THE CURRENT DATE VALUE ON THE	42
43	135	/	DECTAPE]. DISASSEMBLY WILL PROCEED FOR IDENTIFIABLE	43
44	136	/	ELEMENTS; DIRECT BINARY COMPARISON WILL BE USED TO	44
45	137	/	CONFIRM SOURCE FILE AUTHENTICITY. UNTIL THIS IS	45
46	138	/	ACCOMPLISHED, ANY IDEAS REGARDING CODE IMPROVEMENT	46
47	139	/	WILL BE DEFERRED; HOWEVER, CERTAIN TENTATIVE CHANGES	47
48	140	/	ARE BEING MADE TO THE SOURCE FILE AS COMMENTS TO BE	48
49	141	/	CONSIDERED AFTER ALL RECOVERABLE SOURCE CODE	49
50	142	/	OPERATIONS HAVE CONCLUDED.	50
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1	144	/	31-JAN-2016	-	NEW P?S/8-SPECIFIC TD8E BOOTSTRAP ADDED; THE BINARY	1
2	145	/			RELEASE IS CONFIRMED COMPATIBLE WITH THIS NEWER CODE.	2
3	146	/			[NOTE: SEE SPECIFICS OF THIS CODE ELSEWHERE IN THIS	3
4	147	/			FILE FOR DESCRIPTION OF NEW FEATURES.] [NOTE: IN	4
5	148	/			LIGHT OF THE FATAL FLAW NOW KNOWN TO EXIST IN THIS	5
6	149	/			SPECIFIC RELEASE, THIS BOOTSTRAP MAY EITHER BE	6
7	150	/			WITHDRAWN OR MODIFIED TO CREATE A CUSTOM BOOTSTRAP FOR	7
8	151	/			VARIOUS P?S/8-SPECIFIC PURPOSES SUCH AS SUPPORTING AN	8
9	152	/			ALTERNATE BOOTSTRAP TO DRIVE UNIT 1 IF AT ALL	9
10	153	/			FEASIBLE. IT IS PRESENTLY IN THE SOURCE CODE BECAUSE	10
11	154	/			IT IS A FUNCTIONAL EXAMPLE OF HOW TO BOOT THIS	11
12	155	/			PARTICULAR SYSTEM DECTAPE; ALSO IT CAN BE ASSEMBLED	12
13	156	/			TO BOOT THIS SYSTEM IN ONE LESS WORD THAN THE	13
14	157	/			STANDARD DEC TD8E BOOTSTRAP. [NOTE: ALL OF THESE	14
15	158	/			CONSIDERATIONS ARE MOOT SINCE THIS IS THE MR8E-C ROM	15
16	159	/			SYSTEM WHICH HAS THE EQUIVALENT FUNCTIONALITY BUILT	16
17	160	/			IN; HOWEVER, THIS MAY BE REVISITED WHEN THE P?S/8	17
18	161	/			TD8E NON-ROM SYSTEM IS ATTEMPTED SOME TIME IN THE	18
19	162	/			FUTURE; IT IS HOPED THAT AN EXAMPLE OF THAT SYSTEM IS	19
20	163	/			RECOVERED. OTHERWISE, IT WILL HAVE TO BE REDESIGNED	20
21	164	/			AND RECODED FROM SCRATCH.] FUTURE RELEASES OF THIS	21
22	165	/			FILE WILL REMOVE ALL NON-FUNCTIONAL CODE FRAGMENTS AT	22
23	166	/			THE APPROPRIATE TIME AS PART OF A CODE CLEANUP	23
24	167	/			OPERATION, ETC.	24
25	168					25
26	169					26
27	170	/	05-FEB-2016	-	IMAGE OF BLOCK 0 RECREATED IN SOURCE CODE [CONFIRMED	27
28	171	/			ACCURATE. NOTE: THIS RELEASE IS ACTUALLY DEFECTIVE AS	28
29	172	/			A PRACTICAL MATTER; WORK STARTED ON CORRECTING DEFECTS	29
30	173	/			FOR A FUTURE RELEASE ONCE ENOUGH OF THE CODE IS	30
31	174	/			RECOVERED TO ALLOW APPLICATION TO P?S/8 VERSION 8Z.	31
32	175	/			CERTAIN LOCATIONS WITHIN THE HANDLER ARE INDICATED AS	32
33	176	/			CONTAINING SPECIFIC VALUES THAT ARE A PART OF THE	33
34	177	/			ATTEMPT TO BOOT THE SYSTEM IN ONE TAPE PASS. WHEN	34
35	178	/			THE FATAL FLAW IS CORRECTED, THESE LOCATIONS WILL BE	35
36	179	/			MODIFIED TO REFLECT THEIR INTENDED VALUE.	36
37	180					37
38	181	/	18-FEB-2016	-	WORK COMMENCING ON THE SLURP LOADER WHICH WAS	38
39	182	/			DISCOVERED BY TRIAL AND ERROR. P?S/8 CONVENTIONS HAVE	39
40	183	/			CHANGED OVER THE YEARS FOR PLACEMENT OF THIS PORTION	40
41	184	/			OF THE OPERATING SYSTEM DEVICE-DEPENDENT COMPONENTS.	41
42	185	/			WHEN VERSION 8T WAS RELEASED CIRCA 1974/1975, THE	42
43	186	/			"BIN" SYSTEM PROGRAM [THAT LAUNCHES THE SLURP LOADER	43
44	187	/			WHEN NEEDED], INCLUDED THIS BLOCK WITHIN ITS OWN	44
45	188	/			CORE IMAGE.	45
46						46
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1	190	/	THIS IS POOR DESIGN SINCE THAT MEANS THAT SHOULD A	1
2	191	/	TD8E DECTAPE BE ACCESSED ON A TC01/TC08 [OR OTHER	2
3	192	/	POTENTIAL PROBLEMATIC INTERACTION] OCCUR, THE "BIN"	3
4	193	/	PROGRAM ON THAT SYSTEM WILL LOAD THE INCORRECT SLURP	4
5	194	/	LOADER. WHILE THERE CANNOT BE PROTECTION FROM THIS IN	5
6	195	/	GENERAL [AS CERTAIN SYSTEM CONVENTIONS ONLY CHANGE	6
7	196	/	SLOWLY IF AT ALL], THIS SYSTEM IS SO FAR REMOVED FROM	7
8	197	/	NEWER RELEASES, THE ABILITY TO EVEN ACCESS "BIN" FROM	8
9	198	/	THESE EARLIER RELEASES IS ACTUALLY IMPOSSIBLE [AND WHY	9
10	199	/	THE DECTAPE HAD TO BE SEARCHED MANUALLY TO EVEN LOCATE	10
11	200	/	THE CODE!]. HOWEVER, THE NEWER DESIGN ELIMINATES	11
12	201	/	DEVICE-DEPENDENT CODE FROM THE "BIN" PROGRAM AND	12
13	202	/	PLACES THE COMPONENT ON AN IMPLIED SYSTEM BLOCK.	13
14	203	/	THUS, IF ALL SYSTEMS ARE WITHIN A CLOSE RANGE OF	14
15	204	/	RELEASE, THIS PROBLEM IS ELIMINATED AS THE SLURP	15
16	205	/	LOADER WILL ALWAYS BE READ FROM THE IMPLIED BLOCK ON	16
17	206	/	THE SYSTEM DEVICE LOGICAL UNIT FOR BOOTUP.	17
18	207			18
19	208	/	NOTE: THE '/I' DEVICE HANDLER RELOAD CODE HAS ALSO	19
20	209	/	BEEN LOCATED; THIS CAN BE WORKED ON ONCE THE SLURP	20
21	210	/	LOADER HAS BEEN COMPLETELY DOCUMENTED AT THE SOURCE	21
22	211	/	LEVEL, ETC. [NOTE: FORTUNATELY, WHILE THESE BLOCK	22
23	212	/	ASSIGNMENTS HAVE CHANGED OVER THE YEARS, THE OVERALL	23
24	213	/	FUNCTIONALITY HAS NOT, ALTHOUGH A FEW MINOR	24
25	214	/	ADDITIONAL LIMITATIONS MAY APPLY THAT WILL REQUIRE	25
26	215	/	MODIFICATION FOR FUTURE RELEASES [AS WELL AS FOR USE	26
27	216	/	WITH THE PRESENT RELEASE VERSION 8Z]. ALSO, ANY	27
28	217	/	CHANGES MADE TO OVERCOME THE FATAL FLAW MAY INTERACT	28
29	218	/	WITH THIS ASPECT OF THE OPERATING SYSTEM AND THUS	29
30	219	/	AFFECT CERTAIN SPECIFICS OF THE RELOAD; THE RELOAD	30
31	220	/	CONCEPT INTERACTS THREE SPECIFIC PORTIONS OF THE	31
32	221	/	SYSTEM CONFIGURATION: 1) THE SPECIFICS OF THE SYSTEM	32
33	222	/	DEVICE HANDLER WHICH WILL BE UNDERGOING NECESSARY	33
34	223	/	CHANGES TO OVERCOME THE FATAL FLAW, 2) THE SLURP	34
35	224	/	LOADER SPECIFICS THAT WILL BE MODIED, AND 3) THE	35
36	225	/	RELOAD BLOCK ITSELF AS NECESSARY; ALL THREE MUST BE	36
37	226	/	UPDATED IN A COOPERATIVE EFFORT TO ACHIEVE A WORKABLE	37
38	227	/	RELEASE FOR ANY SYSTEM DEVICE THAT SUPPORTS A SLURP	38
39	228	/	LOADER.] [NOTE: ALL TAPE-BASED SYSTEMS AND RX-BASED	39
40	229	/	SYSTEMS SUPPORT A SLURP LOADER AND FEW ELSE. DEVICES	40
41	230	/	THAT RUN ON VERY FAST OR ONE-CYCLE DMA INTERFACES	41
42	231	/	MUST ALWAYS USE THE VIRTUAL LOADER WHICH IS ALWAYS AN	42
43	232	/	OPTION FOR ALL SYSTEMS [BUT MAY BE DEFAULT ON THESE	43
44	233	/	HIGHER-PERFORMANCE CONFIGURATIONS].]	44
45	234			45
46	235	/	THE REWIND AND UNLOAD ROUTINES ARE NOW ROUGHED OUT,	46
47	236	/	BUT NOT FLESHED OUT YET; THE DEVICE SIZE PATCH IS	47
48	237	/	NOW PRESENT.	48
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1	239	/	25-FEB-2016	-	SLURP LOADER FOR THE TD8E WITH MR8EC ROM NOW RECREATED	1
2	240	/			IN SOURCE FORM [AND CONFIRMED CORRECT BY DIRECT BINARY	2
3	241	/			COMPARISON]. CERTAIN CONVENTIONS OF P?S/8 HAVE	3
4	242	/			CHANGED SLIGHTLY; MINOR EDITING WILL BE NECESSARY TO	4
5	243	/			UPDATE THE SOURCE CODE TO FULL COMPATIBILITY WITH	5
6	244	/			P?S/8 VERSION 8Z. OTHER THAN THE CHANGE IN INTERNAL	6
7	245	/			SYSTEM CONVENTIONS, THE SOURCE CODE APPEARS TO BE	7
8	246	/			COMPLETELY FUNCTION AS INTENDED. MOREOVER, MOST OF	8
9	247	/			THE CODE WILL BE EASILY ADAPTED TO THE P?S/8	9
10	248	/			CONFIGURATION FOR THE TD8E WHEN USED WITH EXTENDED	10
11	249	/			MEMORY [WITHOUT THE MD8E-C ROM]. CERTAIN P?S/8	11
12	250	/			GLOBAL DEFINITIONS ARE TEMPORARILY MODIFIED TO PREVENT	12
13	251	/			ASSEMBLY ERRORS. [NOTE: ONCE THIS SOURCE FILE IS	13
14	252	/			COMPLETED WITH REGARD TO SOURCE CODE RECOVERY, IT WILL	14
15	253	/			BE ARCHIVED; ADDITIONAL CHANGES WILL BE MADE TO	15
16	254	/			ACHIEVE COMPATIBILITY WITH P?S/8 VERSION 8Z.]	16
17	255					17
18	256	/	29-FEB-2016	-	REWIND AND UNLOAD ROUTINE FULLY RECOVERED. CERTAIN	18
19	257	/			DEFINITIONS ARE CONDITIONAL ON VERSION 8T. ALTERNATE	19
20	258	/			DEFINITIONS ARE AVAILABLE TO UPGRADE THE SOURCE CODE	20
21	259	/			TO VERSION 8Z STANDARDS [ASSUMING OTHER CHANGES ARE	21
22	260	/			MADE TO CORRECT ALL RELATED ISSUES].	22
23	261					23
24	262	/			THE ALTERNATE STANDALONE BOOTSTRAP FOR THE TD8E IS NOW	24
25	263	/			SHORTER THAN THE DEC STANDARD TD8E BOOTSTRAP	25
26	264	/			REGARDLESS OF CONFIGURATION. DUE TO THE DISCOVERY OF	26
27	265	/			THE FATAL FLAW IN THE BOOTSTRAP OPERATION OF THE	27
28	266	/			SYSTEM DEVICE HANDLER [AND PROJECTED CHANGES TO BE	28
29	267	/			MADE TO REMEDY THE SITUATION], THIS CODE MAY BE	29
30	268	/			IRRELEVANT TO A CORRECTED RELEASE OF THIS SYSTEM. IT	30
31	269	/			IS ESSENTIALLY EXTRANEIOUS DUE TO THE BUILT-IN	31
32	270	/			BOOTSTRAP FUNCTION IN THE MR8E-C ROM; HOWEVER, THIS	32
33	271	/			BOOTSTRAP PROGRAM MAY BE SIGNIFICANT TO THE TD8E	33
34	272	/			SYSTEM REQUIRING EXTENDED MEMORY [WITHOUT THE MR8E-C	34
35	273	/			ROM]. AS SUCH, IT WAS TESTED ON THE PRESENT [FLAWED]	35
36	274	/			SYSTEM AND RETAINED FOR REFERENCE PURPOSES ONLY.	36
37	275	/			[NOTE: THE FLAWED SYSTEM WILL BOOT UP ONLY IF NO	37
38	276	/			CHANGES ARE MADE TO THE CONTENTS OF BLOCK 0000 UNLESS	38
39	277	/			A KLUDGE IS APPLIED THAT MUST BE CALCULATED EACH AND	39
40	278	/			EVERY TIME A CHANGE IS MADE; THIS IS AN UNACCEPTABLE	40
41	279	/			SITUATION THAT WILL REQUIRE SIGNIFICANT REDESIGN. BY	41
42	280	/			AVOIDING SUCH CHANGES FOR THE PRESENT, IT CAN BE	42
43	281	/			BOOTED BY ANY PROPERLY DESIGNED BOOTSTRAP ROUTINE.]	43
44	282					44
45	283	/	02-MAR-2016	-	SYSTEM DEVICE HANDLER RELOADER CODE ROUGHED OUT. SOME	45
46	284	/			COSMETIC CLEANUP, COMMENT CHANGES AND REPAGINATION.	46
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1	286	/	05-MAR-2016	-	SYSTEM DEVICE HANDLER RELOADER CODE FULLY RECOVERED.	1
2	287					2
3	288	/			SOME COMMENTS ADDED REGARDING FUTURE RELEASES; DESPITE	3
4	289	/			THE AMPLE CODE SPACE ALLOCATED FOR THE RELOADER, SOME	4
5	290	/			TECHNIQUES CAN BE IMPROVED TO BE MORE INLINE WITH THE	5
6	291	/			STANDARD OF PROGRAMMING USED IN OTHER ANALOGOUS CODE	6
7	292	/			FOR OTHER DEVICES. THIS WILL BETTER SERVE AS AN	7
8	293	/			EXAMPLE OF HOW TO WRITE FUTURE RELOADER CODE FOR OTHER	8
9	294	/			DEVICES.	9
10	295					10
11	296	/			INTERACTION WITH THE LOGICAL CONSOLE OVERLAY FOR THIS	11
12	297	/			SYSTEM WILL REQUIRE A LOT OF DISCOVERY AS THE OVERLAY	12
13	298	/			WAS IN A MORE PRIMITIVE FORM IN VERSION 8T; IT WAS	13
14	299	/			ONLY FOR THE VT8E AND NON-CONFIGURABLE. MANY OF THE	14
15	300	/			SAME PRINCIPLES CARRIED OVER IN LATER DEVELOPMENT	15
16	301	/			YEARS, BUT IT IS NOT CLEAR IF ANY OF THIS WORK WAS	16
17	302	/			DONE ON THIS PARTICULAR SYSTEM [AS IT IS AN OPTION OF	17
18	303	/			P?S/8 AND IF NO PARTICULAR IMPROVEMENT IS SOUGHT, IT	18
19	304	/			COULD BE IGNORED. HOWEVER, VIRTUALLY EVERY SUPPORTED	19
20	305	/			DEVICE HAS INCLUDED SOME FORM OF INTERACTION WITH THE	20
21	306	/			LOGICAL CONSOLE OVERLAY; THE TD8E SYSTEM DEVICE	21
22	307	/			SYSTEM DEVICE HANDLER CAN CLEARLY BENEFIT FROM THE	22
23	308	/			KIND OF ENHANCEMENT PROVIDED FOR OTHER DEVICES SUCH AS	23
24	309	/			THE TC01/TC08 DECTAPE.	24
25	310					25
26	311	/			FROM THE RECOLLECTIONS OF CHARLES LASNER, IT WOULD	26
27	312	/			APPEAR SOME MEASURE OF OVERLAY SUPPORT WAS ADDED.	27
28	313	/			HOWEVER, IT IS POSSIBLE THIS IS A PRELIMINARY RELEASE	28
29	314	/			CREATED FOR EXPEDIENT PURPOSES; THE CODE MAY SIMPLY	29
30	315	/			NOT BE PRESENT ON THIS PARTICULAR SYSTEM IMAGE FROM	30
31	316	/			THE RECOVERED DECTAPE. MOREOVER, ESPECIALLY IN LIGHT	31
32	317	/			OF THE FATAL FLAW DISCOVERED IN THE LOGIC OF THE	32
33	318	/			SYSTEM DEVICE HANDLER [PREVENTING MODIFICATION OF	33
34	319	/			BLOCK 0000 WITHOUT SOME EXTERNAL REMEDY APPLIED AFTER	34
35	320	/			THE FACT], IT IS LIKELY THAT AT LEAST SOME PORTION OF	35
36	321	/			THIS SUPPORT NEVER EXISTED BEFORE. A COMPLETED FUTURE	36
37	322	/			RELEASE OF A SYSTEM FOR THE TD8E/MR8E-C ROM SHOULD	37
38	323	/			INCLUDE ALL SUCH IMPROVEMENTS [AS WELL AS RESOLVING	38
39	324	/			THE FATAL FLAW AND RECONCILING CERTAIN SYSTEM	39
40	325	/			DEFINITIONS THAT HAVE CHANGED OVER THE YEARS].	40
41	326	/			CERTAIN OF THESE CHANGES HAVE BEEN DOCUMENTED IN THE	41
42	327	/			SOURCE CODE IN THE FORM OF COMMENTS; HOWEVER,	42
43	328	/			ADDITIONAL WORK IS REQUIRED TO PRODUCE A COMPLETELY	43
44	329	/			FUNCTIONAL SYSTEM AS INTENDED [FOR VERSION 8Z AND	44
45	330	/			COMPATIBILITY WITH THE FUTURE REQUIREMENTS OF VERSION	45
46	331	/			9A WHICH IS CURRENTLY IN DESIGN DEVELOPMENT].	46
47	332					47
48	333	/			DEPENDING ON THE LEAD TIME FOR FURTHER WORK ON THIS	48
49	334	/			CONFIGURATION, A PROVISIONAL SYSTEM MAY BE PRODUCED TO	49
50	335	/			PROVIDE NOMINAL COMPATIBILITY WITH VERSION 8Z TO	50
51	336	/			ALLOW TESTING ON BOTH THE SIMH SIMULATOR AND ALSO ON	51
52	337	/			AVAILABLE HARDWARE.	52
53						53
54						54
55						55
56						56
57						57
58						58

1	339	/	IT IS BECOMING INCREASINGLY CLEAR THAT THIS SPECIFIC	1
2	340	/	CONFIGURATION CANNOT SUPPORT BOOTUP FROM DRIVE UNIT 1.	2
3	341	/	AS SUCH, REFERENCES TO THIS WILL EVENTUALLY BE REMOVED	3
4	342	/	FROM ALL SOURCE CODE. [THE NON-ROM TD8E WITH EXTENDED	4
5	343	/	MEMORY SYSTEM VARIANT COULD STILL SUPPORT SUCH AN	5
6	344	/	EXTENDED BOOTSTRAP; WHERE CODE IS INCLUDED FOR FUTURE	6
7	345	/	USE IN DEVELOPING THAT ALTERNATIVE SYSTEM, DRIVE 1	7
8	346	/	REFERENCES WILL BE RETAINED BUT NOTED THAT THE CODE	8
9	347	/	IS PROVIDED [OR COMMENTED ON] PRESUMING THIS IS	9
10	348	/	POSSIBLE [IF ONLY IN THEORY].	10
11				11
12				12
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15				15
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50				50
51				51
52				52
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56				56
57				57
58				58

1	350	/	P?S/8 SYSTEM EXTERNAL EQUATED SYMBOLS.	1
2	351			2
3	352	/	THE FOLLOWING SYMBOLS ARE SPECIFIC TO VERSION 8Z OF P?S/8 AND ARE SUBJECT TO	3
4	353	/	CHANGE IN ANY OTHER RELEASE. [NOTE: MOST SYMBOLS ARE INHERENTLY STABLE AS	4
5	354	/	CORE DEFINITIONS OF THE SYSTEM; HOWEVER, OTHER SYMBOLS ARE DEPENDENT ON EITHER	5
6	355	/	SPECIFIC SYSTEM COMPONENTS OR SYSTEM GENERATION METHODS WHICH MAY CHANGE	6
7	356	/	CONSIDERABLY OR EVEN BECOME OBSOLETE.]	7
8	357			8
9	358	/	THE SYMBOLS ARE DEFINED IN A PRESCRIBED WAY: THE STARTING AND ENDING SYMBOLS	9
10	359	/	MUST BE WITHIN TWO COMMENTS WHICH START WITH '///'; NO OTHER FORM OF COMMENTARY	10
11	360	/	IS ALLOWED OTHER THAN A SINGLE COMMENT ON EACH DEFINITION LINE.	11
12	361			12
13	362	//	P?S/8 EQUATED SYMBOLS.	13
14	363			14
15	364	7603	ADRLST= 7603 /ZAP ADDRESS LIST.	15
16	365	0043	BLEN= 0043 /SLURP LOADER BOOTSTRAP LENGTH.	16
17	366	7734	BCIF= 7734/ **** 8Z **** 7746 /SLURP LOADER STARTING FIELD LOCATION.	17
18	367	0020	BINADR= 0020 /"%" FILE STARTING BLOCK.	18
19	368	0020	BINIT= 0020 /SLURP INITIALIZE ROUTINE.	19
20	369	0020	BINITLN=0020 /LENGTH OF SLURP INITIALIZE CODE.	20
21	370	0031	BINIT2= 0031 /SECONDARY SLURP INITIALIZE ADDRESS.	21
22	371	7776	BISTRT= 7776 /DEFAULT STARTING ADDRESS IF '/I' OR '/V'.	22
23	372	0020	CALADR= 0020 /CALLING ADDRESS.	23
24	373	0021	CALFLD= 0021 /CALLING FIELD.	24
25	374	1416	CHLOOP= 1416 /CHARACTER INPUT ROUTINE. **** 8U ONLY ****	25
26	375	7615	CNTLST= 7615 /CONTENTS LIST FOR ZAPPED ADDRESSES.	26
27	376	6760	DEVSIZE=6760 /DEVICE SIZE VALUE IN SYSGEN ROUTINE.	27
28	377	0024	DSPBLB= 0024 /BLURB DISPLAY ROUTINE.	28
29	378	0046	ERRTRP= 0046 /ERROR TRAP POINTER.	29
30	379	0077	ESTATUS=0077 /ERROR STATUS.	30
31	380	0023	FILBLB= 0023 /BLURB FILL ROUTINE.	31
32	381	7400	GENAD1= 7400 /SYSIO GENERATION ADDRESS.	32
33	382	7600	HIDWRD= 7600 /SYSTEM HANDLER ID WORD.	33
34	383	4020	IGEN= 4020 /I GENERATION FIELD AND ADDRESS.	34
35	384	0045	IOTRAP= 0045 /I/O TRAP POINTER.	35
36	385	0047	IOWAIT= 0047 /I/O WAIT POINTER.	36
37	386	1305	INSERT= 1305 /CHARACTER INSERT ROUTINE.	37
38	387	0105	L200= 0105 /CONSTANT 0200.	38
39	388	0177	L7= 0177 /CONSTANT 0007.	39
40	389	0056	L77= 0056/ **** 8Z **** 0103 /CONSTANT 0077.	40
41	390	0077	L7000= 0077 /CONSTANT 7000. **** 8U ONLY ****	41
42	391	0064	L7756= 0064/ **** 8Z **** 0065 /CONSTANT 7756.	42
43	392	0040	L7770= 0040 /CONSTANT 7770.	43
44	393	0353	MONSTRT=0353 /KEYBOARD MONITOR STARTUP ADDRESS.	44
45	394	0022	MOVBLB= 0022 /ERROR BLURB MOVE ROUTINE.	45
46	395	0162	PCHLOOP=0162 /POINTER TO CHARACTER INPUT ROUTINE.	46
47	396	1547	REWIND= 1547/ **** 8Z **** 1335 /REWIND/UNLOAD ADDRESS.	47
48	397	7635	SBLOCK= 7635 /SYSTEM LOADER BLOCK ARGUMENT.	48
49	398	7600	SBOOT= 7600 /MONITOR RESTART ADDRESS.	49
50	399	7602	SBTFUN= 7602 /SYSTEM BOOTSTRAP FUNCTION.	50
51	400	7633	SCORE= 7633 /SYSTEM LOADER TRANSFER ADDRESS.	51
52	401	7611	SCRSIZE=7611 /SYSTEM MEMORY SIZE, ETC. WORD.	52
53	402	7610	SDATE= 7610 /SYSTEM LOW-ORDER DATE WORD.	53
54	403	7757	SFILES= 7757 /FILE LIST PASSED HERE.	54
55	404	7634	SFUN= 7634 /SYSTEM LOADER FUNCTION WORD.	55
56				56
57				57
58				58

1	406	7632	SLOAD= 7632	/SYSTEM LOADER ADDRESS.	1
2	407	4420	SLPGEN= 4420	/SLURP LOADER GENERATION FIELD AND ADDRESS.	2
3	408	7777	SLPSA= 7777	/SLURP LOADER STARTING ADDRESS WORD.	3
4	409	2600	SLUPAGE=2600	/SLURP LOADING PAGE.	4
5	410	7607	SOUTFLS=7607	/OUTPUT FILE COUNT PASSED HERE.	5
6	411	7756	SPARM= 7756	/EQUALS PARAMETER PASSED HERE.	6
7	412	7604	SWAL= 7604	/COMMAND-LINE OPTION SWITCHES '/A'-' /L'.	7
8	413	7605	SWMX= 7605	/COMMAND-LINE OPTION SWITCHES '/M'-' /X'.	8
9	414	7606	SWY9= 7606	/COMMAND-LINE OPTION SWITCHES '/Y'-' /9'.	9
10	415	7640	SYSIO= 7640	/SYSTEM I/O ENTRY POINT.	10
11	416	7637	SYSTART=7637	/SYSTEM LOADER STARTUP ADDRESS.	11
12	417	0100	TEMP1= 0100	/TEMPORARY.	12
13	418	1041	TRAPI0= 1041	/ENTRY POINT FOR I/O TRAPS.	13
14	419	1200	TRPERR= 1200	/ENTRY POINT FOR I/O ERROR TRAPS.	14
15	420	1020	TRPGEN= 1020	/TRAP ROUTINE GENERATION FIELD AND ADDRESS.	15
16	421	0000	UNIT= 0000	/KEYBOARD MONITOR UNIT.	16
17	422	1110	WAITIO= 1110	/ENTRY POINT FOR I/O WAIT TRAPS.	17
18	423	4000	WRITE= 4000	/SYSIO WRITE BIT.	18
19	424	7627	ZAPLST= 7627	/REPLACEMENT LIST FOR ZAPPED ADDRESSES.	19
20	425				20
21	426	//	END OF P?S/8 EQUATED SYMBOLS.		21
22	427				22
23	428	/	TD8E SIMPLE DECTAPE INPUT/OUTPUT DEFINITIONS.		23
24	429				24
25	430	0077	SDCODE= 77	/DEVICE CODE FOR TD8E SIMPLE DECTAPE [PRIMARY].	25
26	431				26
27	432	6771	SDSS= SDCODE^10+6001	/SIMPLE DECTAPE SKIP ON SINGLE LINE FLAG.	27
28	433	6772	SDST= SDCODE^10+6002	/SIMPLE DECTAPE SKIP IF TIME ERROR FLAG SET.	28
29	434	6773	SDSQ= SDCODE^10+6003	/SIMPLE DECTAPE SKIP ON QUAD LINE FLAG.	29
30	435	6774	SDLC= SDCODE^10+6004	/SIMPLE DECTAPE LOAD COMMAND REGISTER.	30
31	436	6775	SDLD= SDCODE^10+6005	/SIMPLE DECTAPE LOAD DATA REGISTER.	31
32	437	6776	SDRC= SDCODE^10+6006	/SIMPLE DECTAPE READ COMMAND REGISTER.	32
33	438	6777	SDRD= SDCODE^10+6007	/SIMPLE DECTAPE READ DATA REGISTER.	33
34	439				34
35	440	/	NUMERIC LOAD DEFINITIONS.		35
36	441				36
37	442	7201	NL0001= CLA IAC	/LOAD AC WITH 0001.	37
38	443	/	NL0002= CLA CLL CML RTL	/LOAD AC WITH 0002.	38
39	444	7332	NL2000= CLA CLL CML RTR	/LOAD AC WITH 2000.	39
40	445	7330	NL4000= CLA CLL CML RAR	/LOAD AC WITH 4000.	40
41	446	/	NL7775= CLA CLL CMA RTL	/LOAD AC WITH 7775.	41
42	447	7344	NL7776= CLA CLL CMA RAL	/LOAD AC WITH 7776.	42
43	448	/	NL7777= CLA CMA	/LOAD AC WITH 7777.	43
44	449				44
45	450	/	MISCELLANEOUS EQUATED SYMBOLS.		45
46	451				46
47	452	/	SBOFF= SLUPAGE-SBOOT	/OFFSET VALUE FOR SLURP LOADER.	47
48	453	0000	SYSBLK= 0000	/SYSTEM BOOTSTRAP BLOCK.	48
49	454	0014	SYSNUM= 0014	/SYSTEM ID NUMBER; **** MUST BE UNIQUE. ****	49
50	455	2702	TDSize= 2702	/STANDARD DECTAPE BLOCK LENGTH.	50
51					51
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1	457		/	SLURP FORMAT CONTROL DEFINITIONS.	1
2	458				2
3	459	0000	DA= 0	/DATA WORD.	3
4	460	0001	EF= 1	/END-OF-FILE.	4
5	461	0003	FL= 3	/FIELD SETTING.	5
6	462	0002	OR= 2	/ORIGIN SETTING.	6
7	463		/	ROM ADDRESS DEFINITIONS.	7
8	464				8
9	465				9
10	466	7552	F2SET= 7552	/CONTINUATION OF FUNCTION CALL ADDRESS.	10
11	467	7470	ROMBOOT=7470	/ROM-BASED BOOTSTRAP ROUTINE ADDRESS.	11
12	468	0070	ROMFLD= 70	/ROM LOADS INTO FIELD 7.	12
13	469	7405	ROMGO= 7405	/RETRY ADDRESS AFTER ERRORS.	13
14	470	7677	READLP= 7677	/READ DATA WORD LOOP ADDRESS IN THE ROM.	14
15	471				15
16	472		/	ROM-DEFINED LOCATIONS WITHIN THE SYSTEM DEVICE HANDLER.	16
17	473				17
18	474	7751	BUFADR= 7751	/READ/WRITE BUFFER ADDRESS.	18
19	475	7753	BUFPTR= 7753	/BUFFER POINTER FOR READ AND STORE ROUTINES.	19
20	476	7674	CIFR= 7674	/INTERNAL RETURN ADDRESS FOR WRITING.	20
21	477	7731	DRVUNIT=7731	/DECTAPE DRIVE UNIT.	21
22	478	7704	EQUFUN= 7704	/EQUIVALENCE CHECKSUM ROUTINE.	22
23	479	7750	EQUIMP= 7750	/TEMPORARY FOR EQUIVALENCE CHECKSUM ROUTINE.	23
24	480	7647	GETWRD= 7647	/GET A WORD FROM THE CALLER'S BUFFER ROUTINE.	24
25	481	7752	PAGCNT= 7752	/TRANSFER PAGE COUNT.	25
26	482	7654	PUTWRD= 7654	/PUT A WORD INTO THE CALLER'S BUFFER ROUTINE.	26
27	483	7676	RDQUAD= 7676	/READ A 12-BIT WORD ROUTINE.	27
28	484	7734	SBLK= 7734	/LATEST DECTAPE BLOCK NUMBER.	28
29	485	7733	SCKSUM= 7733	/RUNNING CHECKSUM WORD.	29
30	486	7661	SERR= 7661	/ERROR HANDLER ADDRESS.	30
31	487	7722	SEXIT= 7722	/ROM EXIT WHEN FINISHED.	31
32	488	7732	SFUNCTI=7732	/FUNCTION WORD.	32
33	489	7735	SUNIT= 7735	/DRIVE UNIT FOR LATEST HANDLER CALL.	33
34	490	7730	TRYCNT= 7730	/RETRY COUNTER.	34
35	491	7754	WORDCT= 7754	/WORD COUNTER FOR LATEST DATA BLOCK.	35
36	492	7724	WRQ= 7724	/WRITE QUAD ROUTINE ROM CONTINUATION ADDRESS.	36
37	493	7671	WRQUAD= 7671	/WRITE A 12-BIT WORD ROUTINE.	37
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1	495	/	TD8E BOOTSTRAP METHODS.	1
2	496			2
3	497	/	1. MR8E-C ROM TD8E BOOTSTRAP.	3
4	498			4
5	499	/	ASSUMING THE MR8E-C ROM IS AVAILABLE [WHICH IS A REQUIREMENT OF THIS SYSTEM	5
6	500	/	CONFIGURATION], THE FOLLOWING IS THE SHORTEST WAY TO BOOTSTRAP ANY COMPATIBLE	6
7	501	/	TD8E OPERATING SYSTEM [ON LOGICAL UNIT 0 ONLY] BY THE USE OF AN EXECUTABLE	7
8	502	/	PROGRAM. [USING THE FRONT PANEL SWITCH REGISTER TO BOOTSTRAP THE SYSTEM IS	8
9	503	/	ESSENTIALLY THE MANUAL EQUIVALENT OF THIS PROGRAM.]	9
10	504			10
11	505		NOPUNCH /PREVENT BINARY OUTPUT; THIS SECTION IS FOR	11
12	506		/DOCUMENTATION PURPOSES ONLY.	12
13	507			13
14	508	*0200	*200 /ANYWHERE ON ANY PAGE OF ANY FIELD.	14
15	509			15
16	510	/	CLA /THE AC MUST BE CLEAR.	16
17	511	/	IOF /INTERRUPTS MUST BE OFF.	17
18	512	/	CDF XX /THE DATA FIELD IS IGNORED.	18
19	513	000200 6272	CIF 70 /GOING TO ROM FIELD.	19
20	514	000201* 5602	JMP I .+1; ROMBOOT /START THE BOOTSTRAP WITHIN THE ROM.	20
21	515	000202* 7470		21
22	516			22
23	517	/	NOTE: THIS BOOTSTRAP CAN BE SHORTENED BY ONE INSTRUCTION BY LOADING ANYWHERE	23
24	518	/	ON PAGE 36 [7400-7577]. THE "JMP I" INSTRUCTION CAN THEN BE REPLACED BY	24
25	519	/	"JMP ROMBOOT"; THE ADDRESS POINTER TO "ROMBOOT" BECOMES SUPERFLUOUS.	25
26	520			26
27	521	/	2. STANDARD DEC TD8E BOOTSTRAP.	27
28	522			28
29	523		NOPUNCH /PREVENT BINARY OUTPUT; THIS SECTION IS FOR	29
30	524		/DOCUMENTATION PURPOSES ONLY.	30
31	525			31
32	526	0000	FIELD 0 /LOADS IN FIELD 0.	32
33	527			33
34	528	*7300	*7300 /THIS IS WHERE THE STANDALONE BOOTSTRAP LOADS.	34
35	529			35
36	530	007300* 1312 K1000,	TAD GET/(2000) /GET REVERSE DIRECTION BIT. ALSO CONSTANT	36
37	531		/1000 [ONLY THE HIGH-ORDER FOUR BITS ARE USED].	37
38	532	007301* 4312	JMS GET /LOCATE REVERSE END ZONE.	38
39	533	007302* 4312	JMS GET /LOOK FOR 31 CODE GOING FORWARD.	39
40	534	007303* 6773 RD,	SDSQ /QUAD FLAG UP?	40
41	535	007304* 5303	JMP .-1 /NO, WAIT FOR IT.	41
42	536	007305* 6777	SDRD /YES, READ IT IN.	42
43	537	007306* 3726	DCA I WCT /STORE IT INTO MEMORY.	43
44	538	007307* 2326	ISZ WCT /BUMP TO NEXT; ALL DONE YET?	44
45	539	007310* 5303	JMP RD /NO, KEEP GOING.	45
46	540			46
47	541	/	AT THIS POINT BLOCK 0000 HAS BEEN READ INTO 07360-07557 AND BLOCK 0001 HAS	47
48	542	/	BEEN READ INTO 07600-07777. [NOTE: ONLY OS/8 AND COS-300/310 REQUIRE THE	48
49	543	/	BLOCK 0001 DATA LOAD; EDU-SYSTEM 15/30 AND P?S/8 ONLY REQUIRE BLOCK 0000	49
50	544	/	LOADING INTO MEMORY.] FOR ALL SUPPORTED SYSTEMS, THE OFFSET LOADING INTO	50
51	545	/	FIELD 0 7360-7557 WILL BE EXPECTED AND DEALT WITH ACCORDINGLY.	51
52	546			52
53	547	007311* 5732	JMP I STRT/(STARTIT) /GO START IT UP.	53
54				54
55				55
56				56
57				57
58				58

1	549	/	GET TO DESIRED MARK WINDOW VALUE ROUTINE. [NOTE: INITIALIZED TO SAVE A WORD.]	1
2	550			2
3	551	007312* 2000 GET, 2000	/GET TO DESIRED MARK ROUTINE.	3
4	552	007313* 1300 TAD K1000/(1000)	/ADD ON UNIT [0] AND GO BITS TO DIRECTION BIT	4
5	553		/PASSED IN AC. NOTE: LOW-ORDER BITS IGNORED.	5
6	554	007314* 6774 SDLC	/LOAD COMMAND REGISTER TO START THE TAPE MOVING.	6
7	555	007315* 6771 MRKSRCH,SDSS	/NEW LINE IN?	7
8	556	007316* 5315 JMP .-1	/NO, WAIT FOR IT.	8
9	557	007317* 6776 SDRG	/YES, READ THE COMMAND REGISTER.	9
10	558	007320* 0331 AND K77/(77)	/JUST MARK WINDOW BITS.	10
11	559			11
12	560	/	THE FIRST TIME THIS ROUTINE IS USED, THE REVERSE END ZONE MARK VALUE IS	12
13	561	/	SEARCHED FOR; THE SECOND TIME, THE 31 CODE IS SEARCHED FOR.	13
14	562			14
15	563	007321* 1327 MARKSEL,TAD MRKVALUE	/COMPARE TO DESIRED [END MARK OR 31 CODE].	15
16	564	007322* 7640 SZA CLA	/SKIP IF IT MATCHES.	16
17	565	007323* 5315 JMP MRKSRCH	/GO BACK IF NOT.	17
18	566	007324* 2321 ISZ MARKSEL	/BUMP TO NEXT MARK CODE FOR NEXT TIME.	18
19	567	007325* 5712 JMP I GET	/RETURN TO CALLER.	19
20	568			20
21	569	007326* 7354 WCT, BOOTBUFFER	/WORD COUNT AND STORE POINTER.	21
22	570			22
23	571	007327* 7756 MRKVALU, -22; -31	/REVERSE END AND 31 CODE MARK CONSTANTS.	23
24	572	007330* 7747		24
25	573	007331* 0077 K77, 77	/CONSTANT 0077.	25
26	574	007332* 7400 STRT, STARTIT	/WHERE TO START THE BOOT CODE READ IN.	26
27	575			27
28	576	007333* 0000 ZBLOCK 7354-.	/EMPTY SPACE.	28
29	577			29
30	578	7354 BOOTBUF=.	/DECTAPE BLOCK 0000 CONTENTS START LOADING HERE.	30
31	579			31
32	580	007354* 0000 ZBLOCK 4	/THESE LOCATIONS ARE LOADED WITH PRE-DATA WORDS.	32
33	581			33
34	582	7360 BLKDATA=.	/THE DATA WORDS FROM BLOCK 0000 LOAD HERE.	34
35	583			35
36	584	007360* 0000 ZBLOCK 7400-.	/THE FIRST 0020 WORDS OF BLOCK 0000 LOAD HERE.	36
37	585			37
38	586	7400 STARTIT=.	/THE REST OF THE WORDS IN BLOCK 0000 LOAD	38
39	587	/	HERE. CONTROL IS TRANSFERRED HERE AFTERWARDS.	39
40	588	/	ALL /COMPATIBLE OPERATING SYSTEMS NEED TO BE	40
41	589	/	AWARE OF THIS OFFSET LOADING AND PROVIDE A	41
42	590		/SYSTEM-SPECIFIC PROCESS TO CONTINUE THE OVERALL	42
43	591		/OPERATING SYSTEM LOADING PROCESS.	43
44				44
45				45
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52				52
53				53
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1	593	/	NOTES:	1
2	594			2
3	595	/	1) THE STANDARD DEC TD8E BOOTSTRAP ALSO LOADS THE CONTENTS OF BLOCK 0001 INTO	3
4	596	/	07600-07777; THIS IS ONLY MEANINGFUL TO OS/8 AND COS-300/310.	4
5	597			5
6	598	/	2) THE MR8E-C ROM BOOTSTRAP FUNCTIONS IDENTICALLY TO THE STANDARD DEC TD8E	6
7	599	/	BOOTSTRAP; CLEARLY THE STANDARD DEC TD8E BOOTSTRAP IS THE TEMPLATE FOR THE	7
8	600	/	DESIGN OF THE ROM-BASED ROUTINES.	8
9	601			9
10	602	/	3) THE STANDARD DEC TD8E BOOTSTRAP IS NOT WELL DEVELOPED. IT CAN BE MADE	10
11	603	/	MORE ROBUST BY CHANGING THE "K1000" SYMBOLIC REFERENCE TO THE WORD AT	11
12	604	/	"GET+1" [WHICH WOULD MAKE IT SELF-REFERENTIAL]. THE ONLY REQUIREMENT OF	12
13	605	/	THE CONSTANT USED IS THE HIGH-ORDER FOUR BITS ARE 0, 0, 1, 0; AS WRITTEN,	13
14	606	/	EITHER LOCATION ACCOMPLISHES THIS; ONCE THIS CHANGE IS MADE, ADDITIONAL	14
15	607	/	CHANGES ARE POSSIBLE TO MAKE THIS PROGRAM MORE ROBUST [BY ITSELF, THIS	15
16	608	/	CHANGE IS MERELY INNOCUOUS].	16
17	609			17
18	610	/	4) ASSUMING 3) ABOVE HAS BEEN PERFORMED, THE INSTRUCTION AT 07300 CAN BE	18
19	611	/	CHANGED TO "CLA CLL CML RTR" WHICH LOADS THE ACCUMULATOR WITH 2000 [AKA	19
20	612	/	"NL2000"]. THIS CHANGE HAS THE ADDED BENEFIT OF ENSURING THE PROPER	20
21	613	/	CONTENTS OF THE AC AS THE PROGRAM STARTS, WITHOUT RELIANCE ON POWER CLEAR	21
22	614	/	CONSIDERATIONS [WHICH COULD BE IMPROPERLY CARRIED OUT BY USERS UNFAMILIAR	22
23	615	/	WITH THE PROCESS].	23
24	616			24
25	617	/	5) ASSUMING 3) AND 4) ABOVE HAVE BEEN PERFORMED, THE CONTENTS OF "GET" BECOME	25
26	618	/	UNIMPORTANT. THIS ALLOWS THE FRONT PANEL OPERATIONS NEEDED TO TOGGLE IN	26
27	619	/	THIS CODE TO BE ONE OPERATION SHORTER [THE PREVIOUS VALUE CAN BE LEFT ON	27
28	620	/	THE FRONT PANEL SWITCHES THEN PRESS DEPOSIT TWICE].	28
29	621			29
30	622	/	IT IS CLEAR THIS BOOTSTRAP WAS NOT OPTIMIZED AS IS USUALLY THE CASE WITH OTHER	30
31	623	/	DEC BOOTSTRAP CODING; HOWEVER, IT IS KNOWN THAT AT THE TIME THE TD8E WAS FIRST	31
32	624	/	PRODUCED, THERE WAS WIDESPREAD PROGRAMMER APATHY OR EVEN OPEN DISDAIN FOR THE	32
33	625	/	DEVICE. THE NON-ROM VERSION OF A DEVICE HANDLER FOR OS/8 TO SUPPORT THE TD8E	33
34	626	/	[THAT REQUIRES AT LEAST 12K OF MEMORY] IS DOCUMENTED IN SURVIVING OS/8 SOURCE	34
35	627	/	FILES AS "THE TD8E PROBLEM". THIS ALSO IMPACTED NEGATIVELY ON SUBSEQUENT	35
36	628	/	DEVICE HANDLERS ALSO REQUIRING AT LEAST 12K OF MEMORY; THE HANDLERS FOR THESE	36
37	629	/	NEWER DEVICES WERE PRAGMATICALLY WRITTEN TO SUPERFICIALLY MIMIC QUIRKS OF THE	37
38	630	/	TD8E HANDLER. THIS BECAME NECESSARY TO ALLOW CERTAIN PROGRAMS TO HANDLE "THE	38
39	631	/	TD8E PROBLEM" GENERICALLY EVEN IF THIS REQUIREMENT WAS DETRIMENTAL TO THE	39
40	632	/	HANDLERS FOR THESE OTHER DEVICES.	40
41	633			41
42	634	/	IT WAS SUGGESTED TO THE PDP-8 PROGRAMMING STAFF THAT OS/8 BE REWRITTEN TO	42
43	635	/	CONFORM TO THE EMERGING P?S/8 MEMORY MANAGEMENT STANDARD USED IN ALL P?S/8	43
44	636	/	RELEASES FROM THAT ERA TO THE PRESENT; THEY CONSIDERED IT A GOOD IDEA BUT TOO	44
45	637	/	MUCH WORK TO BOTHER WITH. AS SUCH, THE RELEVANT RELEASES OF OS/8 BASIC AND	45
46	638	/	FORTRAN IV HAVE WEAKNESSES WITHIN THEM DIRECTLY TRACEABLE TO THIS "PROBLEM".	46
47				47
48				48
49				49
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1	640	/	3. P?S/8 TD8E BOOTSTRAP.	1
2	641			2
3	642	/	THIS BOOTSTRAP IS ONLY FOR P?S/8 SYSTEMS BASED ON THE TD8E DECTAPE CONTROLLER.	3
4	643	/	THE STANDARD DEC TD8E BOOTSTRAP WILL ALSO WORK; HOWEVER, IT CANNOT ACCESS	4
5	644	/	DRIVE UNIT 1 AS AN ALTERNATE BOOTUP DEVICE CHOICE. [NOTE: THE CODE PRESENTED	5
6	645	/	HERE DOES NOT ACTUALLY BOOTUP DRIVE UNIT 1; IT IS HOPED THAT FUTURE RELEASES	6
7	646	/	OF P?S/8 FOR TD8E WILL BE COMPATIBLE WITH THIS CODE WITH SUITABLE CHANGES MADE	7
8	647	/	THAT WILL SUPPORT DRIVE UNIT 1.] [NOTE: SUPPORT FOR DRIVE UNIT 1 BOOTUP IN	8
9	648	/	THE P?S/8 TD8E/MR8E-C CONFIGURATION IS CURRENTLY UNDER DEVELOPMENT; IT MAY	9
10	649	/	PROVE IMPOSSIBLE GIVEN THE NEED TO SOLVE THE FATAL FLAW IN THE PRESENT CODE AS	10
11	650	/	WAS RECOVERED FROM THE SPECIFIC DECTAPE IN QUESTION. THIS BOOTSTRAP MAY	11
12	651	/	BECOME WORTHLESS AS THE FATAL FLAW IS REMOVED, OR PERHAPS MIGHT BE USEFUL	12
13	652	/	WHERE IT IS POSSIBLY NEEDED, NAMELY ON THE TD8E CONFIGURATION WITH 8K OR MORE	13
14	653	/	MEMORY WITHOUT THE MR8E-C ROM; FOR THE PRESENT IT IS RETAINED FOR REFERENCE	14
15	654	/	PURPOSES ONLY.]	15
16	655			16
17	656	/	THE STANDARD DEC TD8E BOOTSTRAP IS SELFISHLY DESIGNED TO ALLOW OS/8 BOOTUP	17
18	657	/	EASIER AT THE EXPENSE OF OTHER SYSTEMS COMPATIBLE WITH THIS CONVENTION; AS	18
19	658	/	SUCH, IT IS NEEDLESSLY INFLEXIBLE. [NUANCES OF OTHER SYSTEMS SUCH AS MULTIPLE	19
20	659	/	DRIVE UNIT SUPPORT CANNOT BE IMPLEMENTED.]	20
21	660			21
22	661	/	THE STANDARD DEC TD8E BOOTSTRAP LOADS THE CONTENTS OF BLOCK 0001 INTO	22
23	662	/	07600-07777 WHERE OS/8 [AND COS-300/310] REQUIRE THIS; THE CONTENTS OF BLOCK	23
24	663	/	0000 ARE LOADED INTO 07360-07557 WHICH IS OF NO PARTICULAR ADVANTAGE TO ANY	24
25	664	/	SUPPORTED SYSTEM [AND TENDS TO CAUSE NEEDLESS BOOTUP COMPLEXITY]. AS SUCH,	25
26	665	/	FOR ALL SUPPORTED SYSTEMS [INCLUDING OS/8 AND COS-300/310], WHATEVER ASPECT OF	26
27	666	/	BLOCK 0000 REQUIRING PROPER PAGE ALIGNMENT [GENERALLY TO 7600-7777, AND IN THE	27
28	667	/	CASES OF OS/8 AND COS-300/310 INTO FIELD 1] MIGHT BE NEEDED, IT IS NOT	28
29	668	/	PROVIDED. IT IS NECESSARY TO COPY THE CONTENTS OF BLOCK 0000 TO THE PROPER	29
30	669	/	MEMORY LOCATIONS VIA ADDITIONAL CODING WITHIN EACH SYSTEMS DEVICE HANDLER	30
31	670	/	STRUCTURE.	31
32	671			32
33	672	/	IN THE CASE OF THE P?S/8 TD8E/MR8E-C SYSTEM DEVICE HANDLER, THIS REQUIRES	33
34	673	/	HANDLER CODING TO SUPPORT THE CIRCUMSTANCES OF THIS ODD RELOCATION; MEASURES	34
35	674	/	ARE THEN TAKEN TO UNDO SPECIFIC CORRUPTION CAUSED BY THE MISALIGNMENT.	35
36	675	/	AFTERWARDS, THE HANDLER CODE IS COPIED TO THE INTENDED MEMORY LOCATIONS. DUE	36
37	676	/	TO THE LIMITATIONS OF THE DEC STANDARD TD8E BOOTSTRAP, ONLY DEVICE UNIT 0 CAN	37
38	677	/	BE ACCESSED AT BOOTUP TIME.	38
39	678			39
40	679	/	THE P?S/8-SPECIFIC BOOTSTRAP CORRECTS THESE PROBLEMS BY PROPERLY LOADING THE	40
41	680	/	ENTIRE BLOCK 0000 CONTENTS DIRECTLY INTO 07600-07777 AS INTENDED. THIS WILL	41
42	681	/	ALLOW FUTURE RELEASES TO POTENTIALLY SUPPORT BOOTING TO DRIVE UNIT 1	42
43	682	/	[ASSUMING PROPER REVISIONS ARE MADE TO ALL AFFECTED COMPONENTS]. THE PRESENT	43
44	683	/	VERSION OF THIS BOOTSTRAP IS SEVERAL WORDS SHORTER THAN THE DEC BOOTSTRAP.	44
45	684			45
46	685	/	NOTE: THE BUILT-IN BOOTSTRAP WITHIN THE MR8E-C ROM IS CONSISTENT WITH THE	46
47	686	/	DEC STANDARD TD8E BOOTSTRAP IN ALL RELEVANT WAYS. ONLY THE P?S/8 TD8E	47
48	687	/	BOOTSTRAP HAS THE POTENTIAL TO ACCESS DRIVE UNIT 1 AS DESCRIBED ABOVE.	48
49				49
50				50
51				51
52				52
53				53
54				54
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1	689				NOPUNCH	/PREVENT BINARY OUTPUT; THIS SECTION IS FOR	1
2	690					/DOCUMENTATION PURPOSES ONLY.	2
3	691						3
4	692	*7544		*7544		/STARTING ADDRESS OF BOOT CODE.	4
5	693						5
6	694	/			NOTE: THE FIRST WORD BELOW NEED NOT BE LOADED INTO MEMORY IF THE BOOTSTRAP		6
7	695	/			PROGRAM IS STARTED AT "BUTSTRT" [07562]. IT IS INCLUDED TO CAUSE THE LOWEST		7
8	696	/			LOADING ADDRESS TO BE AN ALTERNATE STARTING ADDRESS; THIS WILL MINIMIZE THE		8
9	697	/			INFORMATION NEEDED FOR AN AVERAGE USER TO SUCCESSFULLY START P?S/8 FOR THE		9
10	698	/			TD8E. [NOTE: IT IS HOPED THIS BOOTSTRAP WILL BE COMPATIBLE WITH THE P?S/8		10
11	699	/			TD8E CONFIGURATION FOR USE WITH 8K OR LARGER SYSTEMS THAT DO NOT INCLUDE THE		11
12	700	/			MR8E-C ROM.]		12
13	701						13
14	702	007544*	5362	GTMARK,	JMP BUTSTRT	/JUMP TO REAL START.	14
15	703	007545*	1345	GT1000,	TAD GT1000/(1000)	/ADD ON UNIT 0, GO BIT TO PASSED DIRECTION.	15
16	704	007546*	6774		SDLC	/LOAD COMMAND REGISTER TO START THE TAPE MOVING.	16
17	705	007547*	6771	GTSRCH,	SDSS	/NEW LINE IN?	17
18	706	007550*	5347		JMP .-1	/NO, WAIT FOR IT.	18
19	707	007551*	6776		SDRC	/YES, READ THE COMMAND REGISTER.	19
20	708	007552*	0360		AND GT77/(77)	/JUST MARK TRACK BITS.	20
21	709	007553*	1375	TADGTCM,	TAD GTCMPVALUE	/GET END MARK [OR 40 MARK] COMPARISON VALUE.	21
22	710	007554*	7640		SZA CLA	/SKIP IF IT MATCHES.	22
23	711	007555*	5347		JMP GTSRCH	/GO BACK IF NO MATCH.	23
24	712	007556*	2353		ISZ TADGTCMP	/LOOK FOR 40 CODE SECOND TIME THROUGH.	24
25	713	007557*	5744		JMP I GTMARK	/RETURN TO CALLER.	25
26	714						26
27	715	007560*	0077	GT77,	77	/CONSTANT 0077.	27
28	716	007561*	7577	GTCT,	SBOOT-1	/WORD COUNT AND STORE POINTER.	28
29	717						29
30	718	/			THE BOOTSTRAP STARTS HERE [OR JUMPED TO FROM THE LOWEST LOADING ADDRESS].		30
31	719						31
32	720	007562*	7332	BUTSTRT,	NL2000	/SET REVERSE DIRECTION BIT.	32
33	721	007563*	4344		JMS GTMARK	/REWIND TAPE 0 INTO THE END ZONE.	33
34	722	007564*	4344		JMS GTMARK	/READ FORWARD TO 40 MARK.	34
35	723	007565*	6773	GTLOOP,	SDSQ	/QUAD LINE FLAG UP?	35
36	724	007566*	5365		JMP .-1	/NO, WAIT FOR IT.	36
37	725	007567*	6777		SDRD	/YES, GET A WORD.	37
38	726	007570*	3761		DCA I GTCT	/STORE INTO BOOT AREA [MINUS ONE].	38
39	727	007571*	2361		ISZ GTCT	/DONE ENOUGH?	39
40	728	007572*	5365		JMP GTLOOP	/NO, GO DO ANOTHER ONE.	40
41	729	007573*	6774		SDLC	/YES, STOP THE TAPE.	41
42	730	007574*	7410		SKP	/SKIP TO THE SMA SZA CLA CONSTANT.	42
43							43
44							44
45							45
46							46
47							47
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1	732	007575*	7756	GTCMPVA, -22	/REVERSE END ZONE MARK CONSTANT.	1
2	733	007576*	7740	SMA SZA CLA	/40 CODE MARK CONSTANT; THIS WILL SKIP THE NEXT.	2
3	734					3
4	735			/	THIS IS THE END OF THE BOOTSTRAP CODE TO BE LOADED. THE NEXT LOCATION WILL BE	4
5	736			/	SKIPPED BECAUSE THE CONTENTS ARE COPIED FROM THE BOOTUP DECTAPE MEDIA AND ARE	5
6	737			/	NOT TRUSTWORTHY. [THE CONTENTS OF PART OF THE REVERSE PCC MARK DATA IS NOW	6
7	738			/	LOADED INTO 07577; THIS VALUE IS OFTEN 7777.] THE NEXT LOCATION TO BE	7
8	739			/	EXECUTED IS 07600 WHICH NOW CONTAINS THE CONTENTS OF THE FIRST DATA WORD [AND	8
9	740			/	IS FOLLOWED BY THE REST OF THE DATA WORDS READ IN FROM BLOCK 0000.]	9
10	741					10
11	742	007577*	0000	AT7577, ...	/THIS GETS STORED INTO.	11
12	743					12
13	744			/	THE CODE FALLS THROUGH INTO THE BOOTUP CODE READ IN FROM THE DECTAPE. AS	13
14	745			/	APPROPRIATE, THE SYSTEM DEVICE HANDLER MAY BE MODIFIED TO BOOT TO DRIVE UNIT 1	14
15	746			/	[ASSUMING THIS IS SUPPORTED].	15
16	747					16
17	748		7600	SBOOT= .	/SYSTEM BOOTS HERE.	17
18	749					18
19	750			ENPUNCH	/RESTORE NORMAL BINARY OUTPUT.	19
20	751					20
21	752			/	NOTE: THIS BOOTSTRAP MAY NOT BE FEASIBLE DEPENDING ON THE INTERNALS OF THE	21
22	753			/	P?S/8 TD8E HANDLER WHICH IS UNDERGOING DESIGN CHANGES; IT IS CONCEIVABLE THAT	22
23	754			/	ELEMENTS OF THIS BOOTSTRAP MAY BECOME RELEVANT TO THE NON-ROM CONFIGURATION.	23
24	755			/	HOWEVER, IT HAS BEEN TESTED WITH THE RECOVERED [BUT FLAWED] TD8E/MR8E-C	24
25	756			/	DECTAPE WHICH USES A METHOD OF BOOTUP THAT IS COMPATIBLE WITH THIS BOOTSTRAP	25
26	757			/	[AS WELL AS THE STANDARD DEC TD8E BOOTSTRAP, WHICH IS LONGER].	26
27						27
28						28
29						29
30						30
31						31
32						32
33						33
34						34
35						35
36						36
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58						58

1	759	/	TD8E/MR8E-C DECTAPE SYSTEM DEVICE HANDLER.	1
2	760			2
3	761	0000	FIELD 0 /WHERE THIS LOADS WHEN GENERATED/EXECUTED.	3
4	762			4
5	763	*7400	*GENAD1 /REQUIRED FOR SYSTEM GENERATION PURPOSES.	5
6	764			6
7	765	*7600	RELOC SBOOT /WHERE THE CODE EXECUTES	7
8	766			8
9	767	007600* 4240	SBOOT, JMS SYSIO /CALL SYSTEM I/O ROUTINES TO BOOTUP.	9
10	768	007601* 7600	SBOOT /READ INTO 07600 AND ONWARD.	10
11	769	007602* 2000	SBTFUN, 20^100+... /FUNCTION WORD; LOW-ORDER UNIT BIT MAY BE SET.	11
12	770	007603* 0000	SYSBLK /BLOCK 0000 AND ONWARD.	12
13	771			13
14	772	/	ALTHOUGH SEEMINGLY A SUBROUTINE CALL, THERE IS NO ACTUAL RETURN TO THE CALLER.	14
15	773	/	ONCE INITIATED, DATA TRANSFERS OVERLAY MEMORY; KEY CONTROLLING ADDRESSES AND	15
16	774	/	OTHER PARAMETERS READ IN FROM BLOCK 0000 TAKE CONTROL OF THE OVERALL PROCESS	16
17	775	/	IN MEMORY. THE ACTUAL TRANSFER IS 07600-03577 [USING ADDRESS WRAP-AROUND	17
18	776	/	KNOWN TO CONTINUE IN FIELD 0]. THE FUNCTION WORD IS SET TO READ 20 BLOCKS	18
19	777	/	INTO MEMORY [BLOCKS 0000-0017]. THIS INITIALIZES THE KEYBOARD MONITOR [AND	19
20	778	/	BUILT-IN LINE EDITOR], THE RESIDENT COPY OF THE TFS DIRECTORY AND THE KEYBOARD	20
21	779	/	MONITOR TRANSIENT COMMAND AREA. [NOTE: FINALIZATION OF THESE SYSTEM	21
22	780	/	COMPONENTS IS PERFORMED WITHIN THE KEYBOARD MONITOR.]	22
23	781			23
24	782	/	ONCE COMPLETELY LOADED [AND MOVED BY THE KEYBOARD MONITOR FOR CERTAIN AREAS]	24
25	783	/	THE MEMORY LAYOUT IS AS FOLLOWS:	25
26	784			26
27	785	/	BLOCK[S] MEMORY FUNCTION	27
28	786			28
29	787	/	0000 07600-07777 THE SYSTEM DEVICE HANDLER.	29
30	788			30
31	789	/	0001-0014 00000-02777 THE RESIDENT KEYBOARD MONITOR AND LINE EDITOR.	31
32	790			32
33	791	/	N/A 03000-06777 INTERNAL FILE BUFFER FOR THE LINE EDITOR.	33
34	792			34
35	793	/	0015-0016 07000-07377 CACHED COPY OF THE CURRENT UNIT TFS DIRECTORY.	35
36	794			36
37	795	/	0017 07400-07577 THE TRANSIENT AREA OF THE KEYBOARD MONITOR.	37
38	796			38
39	797	/	NOTE: THE TRANSIENT AREA SERVES SEVERAL PURPOSES. THE CONTENTS AND USAGE	39
40	798	/	DEPENDS ON VARIOUS COMMANDS.	40
41	799			41
42	800	007604* 0000	SWAL, ... /COMMAND-LINE OPTION SWITCHES '/A'-' /L'.	42
43	801	007605* 0000	SWMX, ... /COMMAND-LINE OPTION SWITCHES '/M'-' /X'.	43
44	802	007606* 0000	SWY9, ... /COMMAND-LINE OPTION SWITCHES '/Y'-' /9'.	44
45	803	007607* 2316	SOUTFLS, 2316/... /OUTPUT FILE COUNT ETC. WORD.	45
46	804	007610* 2316	SDATE, 2316/... /SYSTEM DATE WORD.	46
47	805	007611* 4000	SCRSIZE, 4000/... /SYSTEM CORE SIZE.	47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
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1	807	/	NOTES ON OUTPUT FILE COUNT, SYSTEM DATE AND SYSTEM CORE SIZE WORDS:	1
2	808			2
3	809	/	THE OUTPUT FILE COUNT WORD WILL BE REDEFINED IN P?S/8 VERSION 9A. AS OF	3
4	810	/	VERSION 8Z, BITS[7-11] CONTAIN THE COUNT OF OUTPUT FILES IN THE PASSED FILE	4
5	811	/	LIST; HOWEVER, BITS[0-6] MUST BE 0000000. BITS[1-6] WILL BECOME THE	5
6	812	/	HIGH-ORDER SIX BITS OF A REDESIGNED DATE WORD THAT COVERS 512 YEARS DIRECTLY.	6
7	813	/	BIT[0] WILL DETERMINE THE STATUS OF THE FILE AND COMMUNICATIONS AREA OF THE	7
8	814	/	OPTIONAL P?S/8 SHELL [CURRENTLY UNDER DEVELOPMENT]. AS SUCH, ALL CONFORMING	8
9	815	/	PROGRAMS MUST MASK OFF BITS[0-6] TO OBTAIN THE OUTPUT FILE COUNT.	9
10	816			10
11	817	/	THE FORMAT OF THE SYSTEM DATE WORD WILL BE MODIFIED TO FORM AN 18-BIT INTEGER	11
12	818	/	WHEN COMBINED WITH BITS[1-6] OF THE OUTPUT FILE COUNT WORD. THIS WILL REQUIRE	12
13	819	/	ALL PROGRAMS THAT ACCESS THE SYSTEM DATE TO BE REWRITTEN. THIS IS ALSO	13
14	820	/	SCHEDULED FOR IMPLEMENTATION IN P?S/8 VERSION 9A.	14
15	821			15
16	822	/	THE CORE SIZE WORD CONTAINS SEVERAL SETS OF MEMORY AND OTHER STATUS BITS:	16
17	823			17
18	824	/	BITS[0-2] CONTAIN THE PHYSICAL SIZE OF MEMORY OF THE SYSTEM.	18
19	825			19
20	826	/	BITS[3-5] CONTAIN THE MAXIMUM SIZE THE LOGICAL CORE SIZE CAN BE SET TO. THIS	20
21	827	/	COULD DIFFER FROM THE PHYSICAL SIZE FOR SEVERAL REASONS SUCH AS THE PRESENCE	21
22	828	/	OF THE LOGICAL CONSOLE OVERLAY IN THE PHYSICAL MAXIMUM FIELD, OR THE SYSTEM	22
23	829	/	DEVICE HANDLER REQUIRES AN EXTENSION SUCH AS IN THE CASE OF THE TD8E SYSTEM	23
24	830	/	THAT REQUIRES 8K OF MEMORY OR MORE [BUT DOES NOT USE THE MR8E-C ROM].	24
25	831			25
26	832	/	BITS[6-8] CONTAIN THE CURRENT LOGICAL CORE SIZE. THIS VALUE CAN BE SET TO	26
27	833	/	ANY VALUE FROM 0 UP THROUGH THAT FOUND IN THE MAXIMUM LOGICAL SIZE STORED IN	27
28	834	/	BITS[3-5]. THIS ALLOWS PROGRAMS TO BE TESTED IN RESTRICTED CONFIGURATIONS TO	28
29	835	/	MIMIC THE LIMITATIONS OF SMALLER MACHINES OR TO PROTECT THE CONTENTS OF MEMORY	29
30	836	/	FOR USER-DEFINED PURPOSES.	30
31	837			31
32	838	/	BIT[9] IS SET IF THE SYSTEM DEVICE HANDLER REQUIRES A DEVICE EXTENSION. THE	32
33	839	/	DEVICE EXTENSION OCCUPIES LOCATIONS 6000-7777 OF THE HIGHEST PHYSICAL FIELD OF	33
34	840	/	MEMORY. [NOTE: CERTAIN SYSTEM PROGRAMS SUCH AS P?S/8 PAL CAN ALLOCATE	34
35	841	/	LOCATIONS 0000-5777 OF THE HIGHEST PHYSICAL FIELD IN CASE THE LOGICAL CONSOLE	35
36	842	/	IS DISABLED AND THE SYSTEM DEVICE HANDLER REQUIRES THE EXTENSION.]	36
37	843			37
38	844	/	BIT[10] IS SET IF THE LOGICAL CONSOLE OVERLAY IS ACTIVE. ENTRY POINTS ARE	38
39	845	/	IMPLIED WITH RESPECT TO SPECIFIC 12-BIT ADDRESSES; THE FIELD IS DEFINED AS ONE	39
40	846	/	FIELD MORE THAN THE FIELD VALUE IN BITS[3-5]. WHILE THIS IS GENERALLY	40
41	847	/	IDENTICAL TO THE FIELD REPRESENTED BY THE PHYSICAL CORE SIZE, THIS METHOD	41
42	848	/	ALLOWS MULTIPLE MEMORY FIELDS TO BE RESERVED FOR UNUSUAL CONFIGURATIONS.	42
43	849			43
44	850	/	BIT[11] IS THE LOGICAL ABORT SWITCH THAT ANY PROGRAM CAN SET TO SIGNAL PROGRAM	44
45	851	/	EXIT BY ABORT [GENERALLY CONTROL-C DETECTED ON THE SYSTEM CONSOLE DEVICE].	45
46	852	/	WHILE COMPATIBLE PDP-8 CONSOLE HARDWARE CAN PROVIDE THIS INFORMATION VIA THE	46
47	853	/	KEYBOARD FLAG BEING RAISED [AND THE KEYBOARD BUFFER CONTAINING THE CONTROL-C	47
48	854	/	CHARACTER], DECMATES ARE NOT CAPABLE OF THIS DUE TO THE INCOMPATIBLE KEYBOARD	48
49	855	/	HARDWARE ON ALL MODELS. BY SETTING BIT[11] AS NECESSARY, FULLY CONFORMING	49
50	856	/	PROGRAMS CAN BE WRITTEN TO COMPLETELY FUNCTION ON ALL PDP-8 AND DECMATE MODELS	50
51	857	/	SUPPORTED. [NOTE: P?S/8 ONLY SUPPORTS "FAMILY OF 8" MODELS WHICH EXPLICITLY	51
52	858	/	EXCLUDE THE PDP-5 AND PDP-8/S; THESE MACHINES ARE NOT COMPATIBLE WITH THE	52
53	859	/	ORIGINAL PDP-8 MODEL RELEASED IN 1965 WHICH IS CONSIDERED THE MINIMUM SYSTEM	53
54	860	/	THAT CAN BE SUPPORTED IN P?S/8 AND ALSO OS/8 WITH SUFFICIENT MEMORY.]	54
55				55
56				56
57				57
58				58

1	862	/	COMES HERE WITH INTERRUPTS OFF TO CONTINUE PROCESSING THE LATEST SYSTEM DEVICE	1
2	863	/	HANDLER CALL.	2
3	864			3
4	865	007612* 6214	SMORE, RDF	4
5	866	007613* 1346	TAD SCIDF/(CIF CDF)	5
6	867	007614* 3325	DCA SFLDXIT	6
7	868	007615* 1640	TAD I SYSIO	7
8	869	007616* 3351	DCA BUFADDRESS	8
9	870	007617* 2240	ISZ SYSIO	9
10	871		/BUMP TO FUNCTION WORD.	10
11	872	/	NOTE: THE FOLLOWING INSTRUCTIONS ARE EXECUTED IN LOCATIONS 07400 ONWARD DURING	11
12	873	/	THE STANDARD DEC TD8E [OR ROM-BASED] BOOTSTRAP; THE EFFECTS OF THIS WILL BE	12
13	874	/	INDICATED IN THE COMMENTS ON THE MISALIGNED CODE AFTER THE NORMAL USAGE BELOW.	13
14	875			14
15	876	007620* 6774	SDLC	15
16	877	007621* 7344	NL7776	16
17	878	007622* 0640	AND I SYSIO	17
18	879	007623* 6775	SDLD	18
19	880	007624* 0327	AND S70/(70)	19
20	881	007625* 1346	TAD SCIDF/(CIF CDF)	20
21	882		/FORM "CIF CDF" INSTRUCTION TO CALLER'S BUFFER	21
22	883		/FIELD. NOTE: THE "CIF" IS EXTRANEIOUS BECAUSE	22
23	884	007626* 3250	DCA GETFLD	23
24	885	007627* 1250	TAD GETFLD	24
25	886	007630* 3255	DCA PUTFLD	25
26	887	/	TAD I SYSTEM	26
27	888	007631* 5336	JMP SMORE2	27
28	889		/CONTINUE THERE.	28
29	890	/	THE FOLLOWING IS WHAT OCCURS WHEN THE ABOVE WORDS ARE LOCATED AT 07400 ONWARD	29
30	891	/	BECAUSE OF THE STANDARD DEC TD8E [OR ROM-BASED] BOOTSTRAP.	30
31	892	/	*7400	31
32	893	/	/WHERE THESE WORDS EXECUTE.	32
33	894			33
34	895	/	DECTAPE DRIVE UNIT 0 IS STILL MOVING FROM THE LAST BOOTUP DATA TRANSFER.	34
35	896			35
36	897	/	SDLC	36
37	898	/	NL7776	37
38	899	/	AND I SYSIO&177+7420/(5654)	38
39	900	/	SDLD	39
40	901	/	AND S70&177+7420/(5625)	40
41	902	/	TAD S6203&177+7420/(2353)	41
42	903	/	DCA GETFLD&177+7420	42
43	904	/	TAD GETFLD&177+7420	43
44	905	/	DCA PUTFLD&177+7420	44
45	906		/DESTROY "WRQUAD" ROUTINE RETURN INSTRUCTION.	45
46	907	/	THE DECTAPE DRIVE HAS CORRECTLY BEEN STOPPED; HOWEVER, TWO LOCATIONS OF THE	46
47	908	/	HANDLER HAVE BEEN DESTROYED. THIS DAMAGE MUST BE REPAIRED BEFORE THE HANDLER	47
48	909	/	IS SAFE TO MOVE TO THE PROPER LOCATIONS.	48
49	910	/		49
50	911	/	JMP SMORE2&177+7420	50
51	912		/CONTINUE IN DAMAGE RECOVERY ROUTINE.	51
52	913		ZBLOCK 7632-.	52
53			/EMPTY SPACE [IF ANY].	53
54				54
55				55
56				56
57				57
58				58

1	915	/	THE SYSTEM LOADER STARTS HERE. VARIOUS ARGUMENTS ARE STORED AS NECESSARY TO			1
2	916	/	IMPLEMENT A VARIETY OF SPECIFIC TASKS [SUCH AS STARTING A SYSTEM PROGRAM OR			2
3	917	/	IMPLEMENTING SYSTEM PROGRAM OVERLAYS].			3
4	918					4
5	919	/	THE DEFAULT CONTENTS WILL SAVE ALL OF FIELD 0 IN BLOCKS 0020-0057 [WHICH IS			5
6	920	/	ALSO THE BLOCKS OF THE TWO FILES "%" [0020-0037] AND "\$" [0040-0057]			6
7	921	/	RESPECTIVELY. THIS DEFINES HOW CORE-IMAGES ARE HANDLED WITHIN P?S/8 WHEN			7
8	922	/	USING SUCH PROGRAMS AS "GET", "START", "ODT" AND "BSAVE". THE CORE IMAGE MAY			8
9	923	/	ALSO BE MOVED TO TWO DESIGNATED TINY FILE SYSTEM [TFS] FILES TO DEFER SUCH			9
10	924	/	WORK TO A LATER TIME WHEN THE TWO FILES ARE RESTORED. WHEN THE "GET" COMMAND			10
11	925	/	[OR EQUIVALENT] IS USED, THE PROGRAM STARTING ADDRESS IS 07632 TO IMPLEMENT			11
12	926	/	THIS DEFAULT ACTION.			12
13	927					13
14	928	007632* 4240 SLOAD, JMS SYSIO	/SYSTEM LOADER ENTRY POINT; DEFAULT VALUES			14
15	929		/SAVE ALL OF FIELD 0 THEN RESTART.			15
16	930	007633* 0000 SCORE, 0000	/BUFFER STARTING ADDRESS; DEFAULT VALUE			16
17	931		/IS LOCATION 00000.			17
18	932	007634* 4000 SFUN, 00^100+WRITE+.-.	/FUNCTION WORD; DEFAULT IS WRITE 40 BLOCKS FROM			18
19	933		/THE FIELD 0 BUFFER [LOCATIONS 00000-07777].			19
20	934		/AS REQUIRED, THE LOGICAL UNIT BITS WILL BE			20
21	935		/FILLED IN TO INDICATE THE BOOTUP UNIT.			21
22	936	007635* 0020 SBLOCK, BINADR	/STARTING BLOCK NUMBER; DEFAULT IS 0020.			22
23	937	007636* 5637 JMP I .+1/(SYSTART)	/START AT THE INDICATED STARTING ADDRESS.			23
24	938	007637* 7600 SYSTART, SBOOT	/STARTING ADDRESS; DEFAULTS TO THE SYSTEM			24
25	939		/RESTART ADDRESS.			25
26	940					26
27	941	/	THE DEVICE HANDLER I/O ROUTINES ENTER HERE. THE CALLER MUST SET THE DATA			27
28	942	/	FIELD TO THE CALLING FIELD; THE CONTENTS OF THE AC AND LINK ARE IGNORED WHEN			28
29	943	/	ENTERED. UPON RETURN THE AC WILL BE CLEAR; THE LINK CONTENTS ARE GENERALLY			29
30	944	/	NOT MEANINGFUL; HOWEVER, MOST P?S/8 SYSTEM DEVICE HANDLERS ALSO CLEAR THE LINK			30
31	945	/	[IF POSSIBLE].			31
32	946					32
33	947	/	SYSTEM DEVICE HANDLER ENTRY POINT.			33
34	948					34
35	949	/	THIS ABSOLUTELY *MUST* BE AT LOCATION 07640.			35
36	950					36
37	951	IFNZRO SZA CLA-. <ERROR .>	/ENSURE ENTRY POINT MATCHES "SZA CLA" VALUE.			37
38	953					38
39	954	007640* 2530 SYSIO, 2530/MONSTR	/SYSTEM HANDLER ENTRY POINT. INITIALIZED FOR			39
40	955		/THE BOOTSTRAP TO STARTUP THE KEYBOARD MONITOR.			40
41	956	007641* 7301 CLA CLL IAC	/CLEAR THE LINK AND SET AC TO 0001.			41
42	957	007642* 6000 S6000, SKON	/SKIP THE NEXT IF INTERRUPTS ARE ON AND			42
43	958		/ALSO TURN THE INTERRUPT SYSTEM OFF.			43
44	959	007643* 7001 IAC	/AC=1 IF INTERRUPTS WERE ON, AC=2 IF			44
45	960		/INTERRUPTS WERE OFF.			45
46	961	007644* 1242 TAD S6000/(IOF-2)	/NOW HAVE "IOF" INSTRUCTION IF INTERRUPTS WERE			46
47	962		/OFF OR "ION" INSTRUCTION IF INTERRUPTS WERE ON			47
48	963		/BEFORE THE SYSTEM DEVICE HANDLER WAS CALLED.			48
49	964	007645* 3324 DCA SINTXIT	/STORE IN LINE FOR EXIT LATER.			49
50	965	007646* 5212 JMP SMORE	/CONTINUE THERE.			50
51						51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	967	/	SUPPORT ROUTINES FOR THE ROM-BASED CODE; THIS CODE CANNOT BE MOVED.	1
2	968			2
3	969	007647* 0000	GETWRD, .-. /GET A WORD FROM THE CALLER'S BUFFER ROUTINE.	3
4	970	007650* 6201	GETFLD, .-.^10+CDF /WILL BE "CIF CDF XX" WHERE XX IS THE	4
5	971		/DATA FIELD OF THE CALLER'S BUFFER.	5
6	972	007651* 1753	TAD I BUFPTR /GET A WORD FROM THE BUFFER.	6
7	973	007652* 6272	CIF ROMFLD /GOING TO ROM FIELD.	7
8	974	007653* 5647	JMP I GETWRD /RETURN TO CALLER.	8
9	975			9
10	976	007654* 7705	PUTWRD, READLP+6 /PUT A WORD INTO THE CALLER'S BUFFER ROUTINE.	10
11	977		/THIS WORD IS INITIALIZED FOR DIRECT OVERLAY	11
12	978		/DURING THE BOOTUP OPERATION [PRESENTLY].	12
13	979	007655* 6201	PUTFLD, .-.^10+CDF /WILL BE "CIF CDF XX" WHERE XX IS THE	13
14	980		/DATA FIELD OF THE CALLER'S BUFFER.	14
15	981	007656* 3753	DCA I BUFPTR /PUT A WORD INTO THE BUFFER.	15
16	982	007657* 6272	CIF ROMFLD /GOING TO ROM FIELD.	16
17	983	007660* 5654	JMP I PUTWRD /RETURN TO CALLER.	17
18	984			18
19	985	/	ROM-DETECTED DECTAPE ERRORS COME HERE.	19
20	986			20
21	987	007661* 7300	SERR, CLA CLL /ENSURE LINK IS CLEAR FOR RETRIES.	21
22	988	007662* 7402	HLT /HALT ON ALL ERRORS.	22
23	989	007663* 6272	CIF ROMFLD /GOING TO ROM FIELD.	23
24	990	007664* 5670	JMP I SRMG0/(ROMGO) /RETRY THE LATEST BLOCK.	24
25	991			25
26	992	/	CALL HANDLING CONTINUES HERE.	26
27	993			27
28	994	/	BY PASSING THE LOGICAL UNIT BIT DIRECTLY TO THIS PART OF THE ROM SETUP CODE,	28
29	995	/	THE RETRY COUNTER SUPPORT IS BYPASSED [ALLOWING THE RETRY COUNTER WORD TO BE	29
30	996	/	REPURPOSED]. [LOOK IN THE OS/8 HANDLER FOR HOW TO INTERACT WITH THE RETRY	30
31	997	/	COUNTER; THIS MAKES TAPE OPERATIONS LESS RELIABLE COMPARED TO P?S/8.]	31
32	998			32
33	999	007665* 1335	SMORE3, TAD SUNIT /GET LOGICAL DRIVE UNIT.	33
34	1000	007666* 6272	CIF ROMFLD /GOING TO ROM FIELD.	34
35	1001	007667* 5730	JMP I SF2SET5/(F2SET+5) /CONTINUE SETUP THERE IN THE ROM.	35
36	1002			36
37	1003	/	NOTE: THE FOLLOWING WORD IS SUBJECT TO DESTRUCTION WHEN THE CONTENTS OF BLOCK	37
38	1004	/	0000 ARE LOADED INTO 07360-07557. THE RECOVERY ROUTINE WILL HAVE TO RECREATE	38
39	1005	/	THE PROPER CONTENTS PRIOR TO MOVING THE ENTIRE PAGE CONTENTS FROM 07360-07557	39
40	1006	/	TO 07600-07777; THIS CAN BE AVOIDED BY ONE OF SEVERAL METHODS:	40
41	1007			41
42	1008	/	1) REWORK THE CODE NORMALLY LOADED AT 07620 TO AVOID THE DESTRUCTION [AND	42
43	1009	/	HOPEFULLY NOT MERELY CAUSE SOME ALTERNATE DESTRUCTION].	43
44	1010			44
45	1011	/	2) REPURPOSE THE LOCATION TO BECOME AN UNIMPORTANT VALUE [OR COMPLETELY	45
46	1012	/	UNREFERENCED] THUS THE DESTRUCTION WOULD BECOME SUPERFLUOUS.	46
47	1013			47
48	1014	007670* 7405	SRMG0, ROMGO /POINTER TO "ROMGO" ROUTINE TO START SEARCH.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1016	/	NOTE: THE PREVIOUS FOUR WORDS MAY BE REWORKED INTO THE FOLLOWING:			1
2	1017					2
3	1018	/S70, 70			/CONSTANT 0070.	3
4	1019	/SR0MGO, ROMGO			/POINTER TO "ROMGO" ROUTINE TO START SEARCH.	4
5	1020					5
6	1021	/ ZBLOCK 1			/EMPTY SPACE THAT MAY BE USEFUL.	6
7	1022					7
8	1023	/ ZBLOCK 1			/EMPTY SPACE THAT CAN BE HARMLESSLY DESTROYED.	8
9	1024					9
10	1025	007671* 0000	WRQUAD, .-. .		/WRITE A 12-BIT WORD TO THE DECTAPE ROUTINE.	10
11	1026					11
12	1027	/			THE ACTUAL CODE IS IN THE ROM; THIS IS USED ONLY TO ESTABLISH THE PROPER	12
13	1028	/			SUBROUTINE LINKAGE AS IT IS CALLED FROM SEVERAL PLACES.	13
14	1029					14
15	1030	007672* 6272	CIF ROMFLD		/GOING TO THE ROM FIELD.	15
16	1031	007673* 5324	JMP WRQ		/CONTINUE THERE.	16
17	1032					17
18	1033	/			COMES BACK HERE FROM THE ROM-BASED CODE THAT CARRIES OUT THE SUBROUTINE CODE.	18
19	1034					19
20	1035	007674* 6272	CIFR, CIF ROMFLD		/GOING TO THE ROM FIELD.	20
21	1036	007675* 5671	JMP I WRQUAD		/RETURN TO CALLER.	21
22	1037					22
23	1038	/			NOTE: THE PREVIOUS WORD IS SUBJECT TO DESTRUCTION WHEN THE CONTENTS OF BLOCK	23
24	1039	/			0000 ARE LOADED INTO 07360-07557. THE RECOVERY ROUTINE WILL HAVE TO RECREATE	24
25	1040	/			THE PROPER CONTENTS PRIOR TO MOVING THE ENTIRE PAGE CONTENTS FROM 07360-07557	25
26	1041	/			TO 07600-07777; THIS CAN ONLY BE AVOIDED BY REWORKING THE CODE THAT STARTS UP	26
27	1042	/			AT 07400 [NORMALLY IN 07620 ONWARD] IF POSSIBLE. WHILE THIS IS UNLIKELY, IT	27
28	1043	/			SHOULD BE POSSIBLE TO ALLOW THE RECOVERY ROUTINE TO FIT INTO THE ALLOTED CODE	28
29	1044	/			SPACE AND ALSO REBOOT THE SYSTEM BY AN ALTERNATE METHOD DESCRIBED BELOW WHILE	29
30	1045	/			STILL RESTORING THE PREVIOUS WORD. [CODE SPACE CAN BE OBTAINED BY OTHER	30
31	1046	/			RELATED METHODS IN A VIABLE OVERALL SOLUTION.]	31
32	1047					32
33	1048	007676* 0000	RDQUAD, .-. .		/READ A 12-BIT WORD FROM THE DECTAPE ROUTINE.	33
34	1049	007677* 6773	SDSQ		/QUAD LINE FLAG UP?	34
35	1050	007700* 5277	JMP .-1		/NO, WAIT FOR IT.	35
36	1051	007701* 6777	SDRD		/YES, READ IN THE LATEST 12-BIT WORD.	36
37	1052	007702* 6272	CIF ROMFLD		/GOING TO ROM FIELD.	37
38	1053	007703* 5676	JMP I RDQUAD		/RETURN TO CALLER.	38
39	1054					39
40	1055	007704* 6364	EQUFUN, 6364/.-. .		/EQUIVALENCE DATA WITH CHECKSUM ROUTINE.	40
41	1056	007705* 7040	CMA		/INVERT THE DATA AND ,XOR. THE DATA WORD.	41
42	1057	007706* 3350	DCA EQU TMP		/SAVE IT.	42
43	1058	007707* 1350	TAD EQU TMP		/GET IT BACK.	43
44	1059	007710* 0333	AND SCKSUM		/GET THE CARRY BITS.	44
45	1060	007711* 7041	CIA		/INVERT FOR SUBTRACTION.	45
46	1061	007712* 7104	CLL RAL		/MOVE OVER.	46
47	1062	007713* 1350	TAD EQU TMP		/NOW ADD THE LATEST.	47
48	1063	007714* 1333	TAD SCKSUM		/ADD THE NON-CARRY BITS TO RUNNING SUM.	48
49	1064	007715* 3333	DCA SCKSUM		/STORE BACK.	49
50	1065	007716* 1350	TAD EQU TMP		/GET THE LATEST DATA BACK.	50
51	1066	007717* 7140	CLL CMA		/NOW HAVE CLEAR LINK AND THE LATEST WORD.	51
52	1067	007720* 6272	CIF ROMFLD		/GOING TO ROM FIELD.	52
53	1068	007721* 5704	JMP I EQUFUN		/RETURN TO CALLER.	53
54						54
55						55
56						56
57						57
58						58

1	1070	/	THE ROM CODE RETURNS HERE WHEN ALL TAPE OPERATIONS HAVE FINISHED.	1
2	1071			2
3	1072	007722* 1335	SEXIT, TAD SUNIT /GET LATEST DRIVE UNIT.	3
4	1073	007723* 6774	SDLC /STOP THE TAPE DRIVE.	4
5	1074			5
6	1075	/	RESTORE INTERRUPT STATUS PRIOR TO THE DEVICE HANDLER CALL.	6
7	1076			7
8	1077	007724* 6002	SINTXIT, IOF /OR ION.	8
9	1078	007725* 6203	SFLDXIT, CIF CDF 00 /WILL BE "CIF CDF" TO CALLER'S FIELD.	9
10	1079	/	ISZ SYSIO /BUMP PAST BLOCK ARGUMENT.	10
11	1080	007726* 5640	JMP I SYSIO /RETURN TO CALLER.	11
12	1081			12
13	1082	007727* 0070	S70, 70 /CONSTANT 0070.	13
14	1083		/NOTE: THIS CONSTANT COULD BE MOVED TO ALLOW	14
15	1084		/THE "ISZ SYSIO" INSTRUCTION IN THIS SECTION.	15
16	1085			16
17	1086	/	ALTHOUGH THE ROM-BASED CODE SETS UP A RETRY COUNTER, THE P?S/8 DEVICE HANDLER	17
18	1087	/	DOESN'T USE IT. BY AVOIDANCE OF THAT SECTION OF THE ROM SETUP CODE, THE	18
19	1088	/	LOCATION CAN BE REPURPOSED.	19
20	1089			20
21	1090	7730	TRYCNT= . /USED AS AN ERROR RETRY COUNT IN OTHER SYSTEMS.	21
22	1091			22
23	1092	007730* 7557	SF2SET5, F2SET+5 /POINTER TO ROM-BASED SETUP WITHOUT RETRY COUNT.	23
24	1093	007731* 0000	DRVUNIT, 0^4000 /ROM COPIES "SUNIT" HERE [REDUNDANT DATA].	24
25	1094	007732* 0000	SFUNCTI, 0^4000 /FUNCTION WORD BIT[0]; CLEAR=READ, SET=WRITE.	25
26	1095		/[NOTE: ALL OTHER BITS ARE IGNORED.]	26
27	1096	007733* 0000	SCKSUM, 0/. . /RUNNING CHECKSUM WORD.	27
28	1097	007734* 0000	SBLK, 0000 /LATEST DECTAPE BLOCK.	28
29	1098	007735* 0000	SUNIT, 0^4000 /DRIVE UNIT FOR LATEST FUNCTION IN BIT[0].	29
30	1099			30
31	1100	/	COMES HERE TO CONTINUE THE HANDLER CALL.	31
32	1101			32
33	1102	007736* 1640	SMORE2, TAD I SYSIO /GET THE FUNCTION WORD AGAIN.	33
34	1103		/NOTE: THE PREVIOUS WORD MAY BE MOVABLE.	34
35	1104	007737* 7010	RAR /UNIT BIT TO LINK.	35
36	1105	007740* 7210	CLA RAR /UNIT BIT NOW IN AC[0].	36
37	1106	007741* 3335	DCA SUNIT /SET THE DRIVE UNIT FOR ROM.	37
38	1107	007742* 2240	ISZ SYSIO /BUMP PAST FUNCTION WORD.	38
39	1108	007743* 1640	TAD I SYSIO /GET STARTING BLOCK NUMBER FOR TRANSFER.	39
40	1109	007744* 3334	DCA SBLK /STORE IT.	40
41	1110	007745* 2240	ISZ SYSIO /BUMP PAST BLOCK ARGUMENT.	41
42	1111		/NOTE: THE PREVIOUS WORD MAY BE REMOVABLE.	42
43	1112	007746* 6203	SCIDF, CIF CDF 00 /GOING TO FIELD 0; ALSO CONSTANT 6203.	43
44	1113	007747* 5265	JMP SMORE3 /CONTINUE THERE.	44
45	1114			45
46	1115	/	NOTE: THE "JMP SMORE3" ABOVE COULD BE REMOVED; THE CODE AT "SMORE3" COULD BE	46
47	1116	/	PLACED HERE IF OTHER CHANGES THROUGHOUT THE HANDLER ARE MADE.	47
48	1117			48
49	1118	/	TAD SUNIT /GET LOGICAL DRIVE UNIT.	49
50	1119	/	CIF ROMFLD /GOING TO ROM FIELD.	50
51	1120	/	JMP I SF2SET5/(F2SET+5) /CONTINUE SETUP THERE IN THE ROM.	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1122	007750*	0000	EQU TMP, 0/..	/TEMPORARY FOR EQUIVALENCE ROUTINE.	1
2	1123	007751*	7600	BUFADRE, SB00T/..	/READ/WRITE BUFFER POINTER.	2
3	1124	007752*	4000	PAGCNT, 20^200/..	/TRANSFER PAGE COUNT.	3
4	1125	007753*	7753	BUF PTR, ./..	/BUFFER POINTER FOR WORD TRANSFERS.	4
5	1126	007754*	7754	WORDCT, ./..	/WORD COUNTER FOR LATEST BLOCK TRANSFER.	5
6	1127			/ ROMGO	/THIS CAN BE USED BY THE RECOVERY ROUTINE OR HELP	6
7	1128				/AVOID THE NEED TO RECOVER A CONSTANT.	7
8	1129			/	NOTE: THE CODE CAN BE REWORKED TO USE THE ADDRESS CONSTANT ABOVE DIRECTLY.	8
9	1130					9
10	1131	007755*	0000	ZBLOCK 7756-.	/EMPTY SPACE.	10
11	1132					11
12	1133		7756	SPARM= .	/=XXXX PARAMETER PASSED HERE.	12
13	1134					13
14	1135		7757	SFILES= .+1	/FILE LIST PASSED HERE THROUGH 07777.	14
15	1136					15
16	1137			/	THE CONTENTS OF THIS AREA IS THE RECOVERY ROUTINE THAT IS LOADED WITH THE SAME	16
17	1138			/	OFFSET AS THE ROUTINE THAT CAUSED THE DESTRUCTION OF TWO WORDS. THIS IS DUE	17
18	1139			/	TO THE OFFSET APPLIED EARLIER [AT 07400 AND FOLLOWING INSTEAD OF THE NORMAL	18
19	1140			/	07620].	19
20	1141					20
21	1142			/	*7356	21
22	1143				/WHERE THIS CODE IS WHEN OFFSET LOADED.	22
23	1144	007756*	7300	CLA CLL	/CLEAN UP.	23
24	1145	007757*	6774	SDLC	/STOP THE TAPE.	24
25	1146					25
26	1147			/	NOTE: THE PREVIOUS TWO INSTRUCTIONS ARE REDUNDANT AS THE AC IS ALREADY CLEAR	26
27	1148			/	AND THE TAPE WAS STOPPED EARLIER. IF THE CODE THAT STARTS AT 07400 IS CHANGED	27
28	1149			/	TO REMOVE THE "SDLC" THERE, IT WOULD BE NECESSARY TO RETAIN IT HERE.	28
29	1150					29
30	1151	007760*	1355	TAD S7405-20/(ROMGO)	/GET A COPY OF THE DESTROYED VALUE.	30
31	1152	007761*	3250	DCA SROMGO-20	/PUT IT BACK.	31
32	1153	007762*	1356	TAD S5671-20/(JMP I WRQUAD)	/GET A COPY OF THE DESTROYED VALUE.	32
33	1154	007763*	3255	DCA CIFR+1-20	/PUT IT BACK.	33
34	1155	007764*	1753	SMVLUP, TAD I SFROM-20	/GET A WORD.	34
35	1156	007765*	3754	DCA I STO-20	/PUT A WORD.	35
36	1157	007766*	2353	ISZ SFROM-20	/BUMP TO NEXT WORD TO MOVE.	36
37	1158	007767*	2354	ISZ STO-20	/BUMP POINTER/COUNTER.	37
38	1159	007770*	5344	JMP SMVLUP-20	/KEEP GOING UNTIL ALL ARE MOVED.	38
39	1160	007771*	5752	JMP I S7600-20	/GO START UP A COMPLETE REBOOT.	39
40	1161					40
41	1162			/	THE CODE ABOVE ACTUALLY EXECUTES AS DEFINED IN THE COMMENTS BELOW.	41
42	1163					42
43	1164			/	TAD S7405&177+7420	43
44	1165			/	DCA GETFLD&177+7420	44
45	1166			/	TAD S5671&177+7420	45
46	1167			/	DCA PUTFLD&177+7420	46
47	1168			/SMVLUP, TAD I SFROM&177+7420	/GET A WORD.	47
48	1169			/	DCA I STO&177+7420	48
49	1170			/	ISZ SFROM&177+7420	49
50	1171			/	ISZ STO&177+7420	50
51	1172			/	JMP SMVLUP&177+7420	51
52	1173			/	JMP I S7600&177+7420	52
53						53
54						54
55						55
56						56
57						57
58						58

1	1175	007772*	7600	S7600,	SBOOT	/SYSTEM RESTART ADDRESS.	1
2	1176	007773*	7360	SFROM,	STARTIT-20	/BLOCK 0000 DATA LOADED HERE.	2
3	1177	007774*	7600	STO,	SBOOT	/BLOCK 0000 DATA NEEDED HERE.	3
4	1178	007775*	7405	S7405,	ROMGO	/ADDRESS OF "ROMGO" ROUTINE.	4
5	1179	007776*	5671	S5671,	JMP I WRQUAD	/"JMP I WRQUAD" INSTRUCTION VALUE.	5
6	1180						6
7	1181			/		NOTE: FUTURE RELEASES OF THIS HANDLER MUST INCLUDE A CALL TO THE SYSTEM DEVICE	7
8	1182			/		HANDLER FOR THE PURPOSE OF RELOADING BLOCK 0001-0017; THE REBOOT AT 07600	8
9	1183			/		["SBOOT"] WILL BE REWORKED TO USE THE ROM-BASED BOOTUP ROUTINE.	9
10	1184						10
11	1185	007777*	0000	ZBLOCK	- .	/EMPTY SPACE.	11
12	1186						12
13	1187		*7600	RELOC		/TURN RELOCATION OFF.	13
14							14
15							15
16							16
17							17
18							18
19							19
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1	1189	/	ADDITIONAL SYSTEM-DEPENDENT P?S/8 COMPONENTS.				1	
2	1190						2	
3	1191	/	UNLIKE OS/8, P?S/8 HAS ADDITIONAL COMPONENTS SPECIFIC TO THE SYSTEM DEVICE				3	
4	1192	/	TYPE. WHEN A SYSTEM IS CREATED, ALL DEVICE-SPECIFIC CODE OVERLAYS AN				4	
5	1193	/	OTHERWISE GENERIC OPERATING SYSTEM AND RELATED COMPONENTS.				5	
6	1194						6	
7	1195	/	REWIND AND UNLOAD ROUTINE.				7	
8	1196						8	
9	1197	/	THE P?S8 KEYBOARD MONITOR SUPPORTS TWO COMMANDS DESIGNATED FOR TAPE-BASED OR				9	
10	1198	/	OTHER SYSTEMS WHERE THE SYSTEM DEVICE CAN BE REWOUND AND/OR UNLOADED; THE				10	
11	1199	/	TEMPLATE FOR THESE COMMANDS IS THE IMPLEMENTATION FOR THE TC01/TC08 DECTAPE				11	
12	1200	/	CONTROLLER. ON THIS CONFIGURATION, COMMANDS CAN BE TYPED THAT ALLOW OVERLAP				12	
13	1201	/	OF TAPE MOTION WITH KEYBOARD MONITOR COMMANDS [CONTROL-R TO REWIND THE TAPE]				13	
14	1202	/	AND UNLOAD THE THE SYSTEM DEVICE ENTIRELY [WHICH REQUIRES WAITING FOR THE				14	
15	1203	/	OPERATION TO COMPLETE].				15	
16	1204						16	
17	1205	/	WHERE POSSIBLE, ATTEMPT TO MIMIC [OR IMPROVE] THIS BEHAVIOR FOR EACH				17	
18	1206	/	CONFIGURATION. THE TD8E CONTROLLER CANNOT ALLOW OVERLAP WITH TAPE REWIND				18	
19	1207	/	BECAUSE THERE IS NO AUTOMATIC END ZONE DETECTION; AS SUCH, THE SPECIFIC				19	
20	1208	/	IMPLEMENTATION DIFFERS IN SOME WAYS. BECAUSE THERE ARE NO BRAKE-RELATED				20	
21	1209	/	ISSUES, THE UNLOAD OPERATION CAN BE INITIATED AND WILL NOT WAIT FOR COMPLETION				21	
22	1210	/	[HOWEVER, THE USER HAS TO STOP THE TAPE FROM INFINITE FLAPPING].				22	
23	1211						23	
24	1212	0000	FIELD	0	/FIELD WHERE THIS CODE IS WRITTEN OUT FROM		24	
25	1213		/WHEN GENERATING A COMPLETE OPERATING SYSTEM.				25	
26	1214						26	
27	1215	*1546	*REWIND-1			/THESE ROUTINES ARE NOT RELOCATED.	27	
28	1216						28	
29	1217	001546	3000	L3000,	3000	/CONSTANT 3000.	29	
30	1218						30	
31	1219	/	THE ROUTINES ARE ENTERED HERE. OTHER THAN COMMON OPERATIONS, EACH COMMAND				31	
32	1220	/	WILL PERFORM ADDITIONAL PROGRAMMING TO COMPLETE THE SPECIFIC TASK.				32	
33	1221						33	
34	1222	001547	4356	REWIND,	JMS	TAPMOVE	/THE AC CONTAINS 0000, START A REWIND	34
35	1223						/ON THE CURRENT SYSTEM DEVICE UNIT.	35
36	1224	001550	1100	TAD	TEMP1	/GET LATEST CHARACTER.		36
37	1225	001551	1064	TAD	L7756/[-"R!300]	/COMPARE TO <^R>.		37
38	1226	001552	7650	SNA	CLA	/SKIP IF OTHER [<^T>].		38
39	1227	001553	5364	JMP	ENDWAIT	/WAIT FOR END ZONE THERE.		39
40	1228						40	
41	1229	/	WE WANT TO UNLOAD THE CURRENT SYSTEM DEVICE UNIT; SELECT THE OTHER DRIVE WHILE				41	
42	1230	/	INITIATING MOTION AS WELL [AND THEN TURN MOTION OFF ON THE OTHER UNIT]. THE				42	
43	1231	/	DESELECTION PROCESS DONE THIS WAY WILL CAUSE THE ORIGINAL UNIT TO FALL OFF THE				43	
44	1232	/	TAKEUP REEL.				44	
45	1233						45	
46	1234	001554	7201	NL0001	/SET UNIT REVERSAL BIT.		46	
47	1235	001555	5373	JMP	DESELECT	/CONTINUE THERE.		47
48							48	
49							49	
50							50	
51							51	
52							52	
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58							58	

1	1237		/	TAPE MOVE IN REVERSE ROUTINE. CALL WITH AC=0000 TO REWIND THE CURRENT SYSTEM	1
2	1238		/	DEVICE UNIT; CALL WITH AC=0001 TO REWIND THE OPPOSITE SYSTEM DEVICE UNIT.	2
3	1239				3
4	1240	001556	0000	TAPMOVE, .-. /REWIND THE TAPE ROUTINE.	4
5	1241	001557	1000	TAD UNIT /ADD ON THE CURRENT LOGICAL DEVICE UNIT.	5
6	1242	001560	7112	CLL RTR /MOVE DESIRED UNIT TO AC[0].	6
7	1243	001561	1346	TAD L3000/(3000) /ADD ON REVERSE DIRECTION AND MOTION BITS.	7
8	1244	001562	6774	SDLC /LOAD THE COMMAND REGISTER.	8
9	1245	001563	5756	JMP I TAPMOVE /RETURN TO CALLER.	9
10	1246				10
11	1247		/	COMES HERE TO WAIT FOR THE END ZONE ON THE CURRENT SYSTEM DEVICE UNIT.	11
12	1248				12
13	1249	001564	6771	ENDWAIT, SDSS /SINGLE-LINE FLAG UP?	13
14	1250	001565	5364	JMP .-1 /NO, WAIT FOR IT	14
15	1251	001566	6776	SDRC /YES, GET THE COMMAND REGISTER CONTENTS.	15
16	1252	001567	0056	AND L77/[77] /JUST MARK WINDOW BITS.	16
17	1253	001570	1064	TAD L7756/[-22] /COMPARE TO END ZONE VALUE.	17
18	1254	001571	7640	SZA CLA /SKIP IF IN END ZONE NOW.	18
19	1255	001572	5364	JMP ENDWAIT /NOT YET, KEEP WAITING.	19
20	1256				20
21	1257		/	WE'RE IN THE END ZONE HAVING REWOUND THE CURRENT SYSTEM DEVICE UNIT. FALL	21
22	1258		/	THROUGH TO THE NEXT ROUTINE WHICH WILL REDUNDANTLY SELECT THE SAME SYSTEM	22
23	1259		/	DEVICE UNIT, THEN CLEAR THE MOTION BIT.	23
24	1260				24
25	1261	1573		DESELEC=. /DESELECT THE CURRENT SYSTEM DEVICE UNIT HERE.	25
26	1262				26
27	1263		/	IF THE AC IS SET TO 0001, THE OTHER SYSTEM DEVICE UNIT WILL BE SELECTED WITH	27
28	1264		/	REVERSE DIRECTION AND MOTION SET. THIS WILL CAUSE THE CURRENT DEVICE UNIT	28
29	1265		/	[WHICH WAS SELECTED FOR REVERSE DIRECTION AND MOTION SET] TO LOSE CONTROL; THE	29
30	1266		/	TAPE WILL COMPLETELY UNLOAD AND NOT STOP TAPE FLAPPING UNTIL THE OPERATOR	30
31	1267		/	STOPS THE TAPE MANUALLY. IF THE AC IS SET TO 0000 THE CURRENT SYSTEM DEVICE	31
32	1268		/	UNIT WILL BE REDUNDANTLY SELECTED FOR REVERSE DIRECTION AND MOTION [WHICH IS	32
33	1269		/	HARMLESS].	33
34	1270				34
35	1271	001573	4356	JMS TAPMOVE /SELECT EITHER THE CURRENT SYSTEM DEVICE	35
36	1272			/LOGICAL UNIT OR THE OTHER SYSTEM DEVICE	36
37	1273			/LOGICAL UNIT [DEPENDING ON HOW WE GOT HERE].	37
38	1274	001574	6776	SDRC /GET THE CURRENT COMMAND REGISTER.	38
39	1275	001575	1077	TAD L7000/[-1000] /REMOVE THE MOTION BIT.	39
40	1276		/	TAD L7000/(7000) /REMOVE THE MOTION BIT. **** VERSION 8Z ****	40
41	1277	001576	6774	SDLC /LOAD THE COMMAND REGISTER TO	41
42	1278			/STOP THE CURRENT DECTAPE UNIT.	42
43	1279	001577	5216	JMP CHLOOP /KEEP GOING THERE.	43
44	1280				44
45	1281		/L7000, 7000	/CONSTANT 7000 **** VERSION 8Z ****	45
46	1282				46
47	1283		/	ZBLOCK .!177-. /EMPTY SPACE [IF ANY]. **** VERSION 8Z ****	47
48	1284				48
49	1285		/	JMP I PCHLOOP/[CHLOOP] /CONTINUE THERE. **** VERSION 8Z ****	49
50					50
51					51
52					52
53					53
54					54
55					55
56					56
57					57
58					58

1	1287	/	DEVICE LOGICAL SIZE.	1
2	1288			2
3	1289	/	THE SYSTEM GENERATION ROUTINES NEED TO KNOW THE OVERALL SIZE OF ANY ONE DEVICE	3
4	1290	/	ADDRESSED BY THE SYSTEM HANDLER; CERTAIN ENTRIES WILL BE MADE IN VARIOUS	4
5	1291	/	STRUCTURES BASED ON THIS VALUE. [AS A PRACTICAL MATTER, THIS SIZE MUST BE	5
6	1292	/	GREATER THAN 1000 BLOCKS TO BE OF MUCH USE.]	6
7	1293			7
8	1294	/	ON VERY LARGE DEVICES, THIS COULD BE AS MUCH AS 10000 BLOCKS [WHICH SHOULD BE	8
9	1295	/	EXPRESSED AS 0000]; THIS IS THE UPPER LIMIT OF ADDRESSABILITY FOR ANY P?S/8	9
10	1296	/	SYSTEM DEVICE HANDLER [BUT MAY NOT BE THE UPPER LIMIT OF THE ACTUAL DEVICE].	10
11	1297			11
12	1298	/	THE P?S/8 SHELL IS CAPABLE OF ADDRESSING MORE STORAGE BLOCKS THAN THE BASIC	12
13	1299	/	P?S/8 SYSTEM THROUGH THE USE OF NON-SYSTEM DEVICE HANDLERS. FOR EXAMPLE, ON	13
14	1300	/	AN RK8E/RK05 SYSTEM CONFIGURED FOR LOGICAL DRIVES 0 THROUGH 3 [WHERE EACH	14
15	1301	/	LOGICAL UNIT IS AN ENTIRE RK05 PACK], THE MAXIMUM POSSIBLE SYSTEM BLOCK	15
16	1302	/	AVAILABLE TO THE BASIC SYSTEM IS 07777 AS DEFINED ABOVE. HOWEVER, THE MAXIMUM	16
17	1303	/	BLOCK AVAILABLE TO THE P?S/8 SHELL ENVIRONMENT WOULD BE 31277, WHERE BLOCKS	17
18	1304	/	10000-31277 ARE ONLY ACCESSIBLE THROUGH NON-SYSTEM HANDLERS AS USED IN THE	18
19	1305	/	SHELL ENVIRONMENT [AS WELL AS THE BASIC SYSTEM PROGRAM "BLKCPY" WHEN IT IS	19
20	1306	/	UPGRADED TO THE NEW SPECIFICATION FOR NON-SYSTEM HANDLERS IN THE VERSION 9A	20
21	1307	/	RELEASE]. AS SUCH, BLOCKS 00000-07777 REPRESENT THE MAXIMUM VALUE FOR THE	21
22	1308	/	BASIC P?S/8 SYSTEM THAT CAN BE CONFIGURED; SMALLER VALUES OF MAXIMUM BLOCK MAY	22
23	1309	/	BE USED IN PRACTICAL CONFIGURATIONS WHERE THE BASIC P?S/8 SYSTEM IS NOT	23
24	1310	/	REQUIRED TO SUPPORT THAT MANY STORAGE BLOCKS. THE SYSTEM MAY BE CONFIGURED	24
25	1311	/	SUCH THAT MORE OF THE AVAILABLE STORAGE IS ALLOCATED TO THE SHELL ENVIRONMENT	25
26	1312	/	AS REQUIRED. [NOTE: IT IS POSSIBLE TO CONFIGURE A BASIC P?S/8 SYSTEM TO ONLY	26
27	1313	/	ADDRESS BLOCKS BLOCKS 00000-00377 OR PERHAPS SOMEWHAT LESS, WITH THE REST OF	27
28	1314	/	THE DEVICE STORAGE CONFIGURED FOR USE IN THE P?S/8 SHELL ENVIRONMENT ONLY.]	28
29	1315			29
30	1316	*6760	*DEVSIZE /OVER SYSTEM GENERATION ROUTINES.	30
31	1317			31
32	1318	006760 2702	TDSIZE /DEFINE OUR DEVICE SIZE.	32
33				33
34				34
35				35
36				36
37				37
38				38
39				39
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1	1320	/	P?S/8 BINARY FILE SLURP LOADER AND BOOTSTRAP FOR THE TD8E WITH MR8E-C ROM.	1
2	1321			2
3	1322	0002	FIELD SLPGEN&70%10 /GENERATION FIELD FOR SLURP LOADER.	3
4	1323			4
5	1324	*4400	*SLPGEN&7600 /GENERATION ADDRESS FOR SLURP LOADER.	5
6	1325			6
7	1326	SLPCORE, NOPUNCH	/FOOL THE ASSEMBLER.	7
8	1327			8
9	1328	0000	FIELD 0 /WHERE IT LOADS LATER.	9
10	1329			10
11	1330	*4400	*SLPCORE /RESET THE ORIGIN BACK IN CASE FIELD CHANGES IT.	11
12	1331			12
13	1332		ENPUNCH /UN-FOOL THE ASSEMBLER.	13
14	1333			14
15	1334	*7600	RELOC SBOOT /WHERE THE SLURP LOADER AND BOOTSTRAP EXECUTES.	15
16	1335			16
17	1336	/	DEVICE DEPENDENT BOOTSTRAP STARTING AT 07600. ALSO USED WITH THE P?S/8	17
18	1337	/	VERSION OF THE PAPER-TAPE "BIN" LOADER [INSTEAD OF THE SLURP LOADER].	18
19	1338			19
20	1339	/	P?S/8 BOOTSTRAP CONVENTIONS ARE AS FOLLOWS:	20
21	1340			21
22	1341	/	1) AC CONTENTS UNIMPORTANT.	22
23	1342			23
24	1343	/	2) LINK CONTENTS UNIMPORTANT.	24
25	1344			25
26	1345	/	3) IN GENERAL, THE INTERRUPT SYSTEM SHOULD BE TURNED OFF; HOWEVER, BECAUSE	26
27	1346	/	INTERRUPTS ARE PARTICULARLY A PROBLEM FOR TD8E-BASED SYSTEMS, THIS	27
28	1347	/	BOOTSTRAP FORCES INTERRUPTS OFF AS THERE IS SUFFICIENT CODE SPACE FOR THE	28
29	1348	/	"IOF" INSTRUCTION. [NOTE: IF THE SYSTEM DEVICE OPERATES BY DMA OR PROGRAM	29
30	1349	/	TRANSFER WITHOUT CRITICAL TIMING, THIS WILL NOT BE A PROBLEM. THE CODING	30
31	1350	/	OF THE P?S/8 KEYBOARD MONITOR AND (EMBEDDED EDITOR) IS SUCH THAT INTERRUPTS CAUSE AN ADDITIONAL SYST	31
32	EM RESTART WITH			32
33	1351	/	INTERRUPTS TURNED OFF.]	33
34	1352			34
35	1353	/	4) THE DATA FIELD MUST BE 0.	35
36	1354			36
37	1355	/	5) ALL WORDS OF THE BOOTSTRAP MUST BE INTACT. AS LONG AS ONLY WELL-BEHAVED	37
38	1356	/	PROGRAMS ARE LOADED, THIS SHOULD NEVER BE A PROBLEM.	38
39	1357			39
40	1358	SBOOT,	/BOOTSTRAP RELOAD STARTS HERE.	40
41	1359	007600* 6002	IOF /TURN INTERRUPTS OFF.	41
42	1360	007601* 7670	B7670, SNA SZL CLA/(-110) /CONSTANT 7670; CLEARS THE AC AND MIGHT SKIP.	42
43	1361	007602* 7470	BROMB00, ROMBOOT/(SNA SZL) /ENTRY POINT INTO THE ROM TO REBOOT; CANNOT	43
44	1362		/SKIP BECAUSE THE AC IS NOW CLEAR.	44
45	1363	007603* 6272	CIF ROMFLD /GOING TO ROM FIELD.	45
46	1364	007604* 5602	JMP I BROMBOOT/(ROMBOOT) /GO TO BOOTUP ROUTINE.	46
47	1365			47
48	1366	7605	SLBTEND=. /ACTUAL END OF BOOTSTRAP-REQUIRED CODE.	48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1368	/	NOTE: AS OF VERSION 8Z, THERE IS AN SPECIFIC UPPER LIMIT ON THE SIZE OF THE	1
2	1369	/	CODE NEEDED TO REBOOT P?S/8 WHEN REMOVED FROM THE SLURP LOADER [TO BE USED	2
3	1370	/	WITH THE P?S/8 VERSIONS OF THE BIN AND RIM LOADERS]. VARIOUS EARLIER RELEASES	3
4	1371	/	MAY HAVE MADE DIFFERENT DEMANDS ON THIS CODE LENGTH [AND VERSION 9A WILL	4
5	1372	/	CHANGE THIS UPPER LIMIT TO ONE WORD LESS]. IT IS NOT EXPECTED THIS NEW	5
6	1373	/	REQUIREMENT WILL AFFECT TOO MANY SYSTEM CONFIGURATIONS; EACH WILL HAVE TO BE	6
7	1374	/	RECODED AS REQUIRED. HOWEVER, THIS CONFIGURATION USES ONE OF THE SMALLEST	7
8	1375	/	BOOTUP SECTIONS; AS SUCH, NO CHANGES WILL BE NEEDED FOR THE P?S/8 TD8E WITH	8
9	1376	/	MR8E-C ROM VARIANT. [NOTE: THE TC01/TC08 SYSTEM, ONCE REASSEMBLED FOR USE	9
10	1377	/	WITH VERSION 9A, IS ONE WORD SHORTER THAN THE NEW LIMITATION. EACH SYSTEM	10
11	1378	/	DEVICE HANDLER WILL BE EVALUATED REGARDING THIS POINT.]	11
12	1379			12
13	1380	/	ALL BOOTSTRAP CODE MUST BE WITHIN 07600-07642 IN P?S/8 VERSION 8Z. [NOTE:	13
14	1381	/	IN VERSION 9A, THIS WILL BE FURTHER RESTRICTED TO 07600-07641. THIS WILL	14
15	1382	/	FACILITATE IMPROVING THE P?S/8 VERSION OF THE PAPER-TAPE "BIN" LOADER BY	15
16	1383	/	ALLOWING ONE ADDITIONAL WORD TO BE AVAILABLE. THIS WILL BE USED TO IMPROVE	16
17	1384	/	THE LOADER; WHEN COMPLETED, IT WILL BE IMMUNE TO INTERNAL DESTRUCTION BY THE	17
18	1385	/	EXECUTION OF ALL PROGRAMS FOR THREE-CYCLE DATA BREAK DEVICES. BY CONTRAST,	18
19	1386	/	NO OTHER VERSION OF THE PAPER-TAPE BIN LOADER IS ROBUST ENOUGH TO PREVENT	19
20	1387	/	DESTRUCTION BY THE EXECUTION OF PROGRAMS FOR AT LEAST ONE OF THE PREVAILING	20
21	1388	/	DEVICES [DF32/RF08, TC58 MAGNETIC TAPE, TC01/TC08 DECTAPE].	21
22	1389			22
23	1390	/	THE SLURP LOADER AND BOOTSTRAP ARE WRITTEN OUT TO A SYSTEM-DESIGNATED BLOCK.	23
24	1391	/	THE P?S/8 "BIN" PROGRAM WILL USE THE ENTIRE BLOCK CONTENTS WHEN LOADING TFS	24
25	1392	/	BINARY FILES; ONLY THE BOOTSTRAP WILL BE USED IN CONJUNCTION WITH THE P?S/8	25
26	1393	/	VERSION OF THE PAPER-TAPE "BIN" AND "RIM" LOADERS. THIS WILL ALLOW RELOAD OF	26
27	1394	/	P?S/8 AS LONG AS THE LOCATIONS FROM 07600-ONWARD ARE NOT DESTROYED BY THE	27
28	1395	/	PROGRAMS LOADED BY BIN [OR RIM].	28
29	1396			29
30	1397	/	ENSURE THIS SIZE CONVENTION IS MET.	30
31	1398			31
32	1399		IFNZRO SBOOT+BBLLEN-.&4000 <ERROR .>	32
33	1401			33
34	1402	/	DETAILED DESCRIPTION OF SLURP FORMAT P?S/8 BINARY FILES.	34
35	1403			35
36	1404	/	EVERY BLOCK OF A SLURP FILE IS INDEPENDENT OF ALL OTHER BLOCKS OTHER THAN ANY	36
37	1405	/	CONSIDERATION OF CURRENT LOADING ORIGIN AND CURRENT LOADING FIELD INHERITED	37
38	1406	/	FROM THE PREVIOUS BLOCK PROCESSED. TO ENSURE PROPER LOADING, EVERY SLURP FILE	38
39	1407	/	CONTAINS INITIAL FIELD 0 AND ORIGIN 0200 SETTINGS; THIS AVOIDS DEPENDENCE ON	39
40	1408	/	ORDER OF LOADING, EVEN IN THE CASE OF MULTIPLE OUTPUT FILES CREATED BY A	40
41	1409	/	SINGLE ASSEMBLY [OR BINARY PAPER-TAPE CONVERSION]. SINCE THIS FIXED SEQUENCE	41
42	1410	/	IS PART OF THE OPENING WORDS OF A BINARY FILE, CONFORMING PROGRAMS CAN	42
43	1411	/	DETERMINE THE FILE IS BINARY. [NOTE: P?S/8 SYSTEM PROGRAMS SUCH AS "FOCAL"	43
44	1412	/	USE BOTH BINARY AND TEXT FILES; BY USING THIS FIXED SEQUENCE AS PART OF A	44
45	1413	/	SIGNATURE CHECK, THE PROGRAM IS ABLE TO DIFFERENTIATE AND PROCESS THE FILES	45
46	1414	/	PASSED IN A MIXED SEQUENCE; FOCAL REQUIRES ALL BINARY FILES TO BE PASSED PRIOR	46
47	1415	/	TO ALL TEXT FILES, ETC.]	47
48	1416			48
49	1417	/	WHILE TFS FILES ARE RESTRICTED TO AT MOST 16 BLOCKS OF DATA, THE SLURP FORMAT	49
50	1418	/	DOES NOT HAVE ANY INHERENT BOUNDARY RESTRICTIONS. FUTURE RELEASES OF P?S/8	50
51	1419	/	WILL SUPPORT THE "BINCON" UTILITY WITH THE ABILITY TO CREATE EXTENDED-LENGTH	51
52	1420	/	BINARY FILES THAT CAN BE CONVERTED TO OS/8 BINARY FORMAT AFTER PROCESSING TO	52
53	1421	/	REMOVE THE PROBLEMS ASSOCIATED WITH THE DEFECTIVE IMPLEMENTATION OF THE DTORG	53
54	1422	/	DIRECTIVE. THIS IN TURN WILL FACILITATE FUTURE METHODS OF P?S/8 SYSTEM	54
55	1423	/	GENERATION.	55
56				56
57				57
58				58

1	1425	/	SLURP LOADER BLOCK FORMAT.	1
2	1426			2
3	1427	/	WITHIN ANY SLURP FORMAT BINARY BLOCK THERE ARE 18 GROUPS OF SEVEN WORDS. ANY	3
4	1428	/	ADDITIONAL WORDS ARE IGNORED.	4
5	1429			5
6	1430	/	SINCE SLURP LOADERS OPERATE AT THE PHYSICAL DEVICE LEVEL, EACH SLURP LOADER	6
7	1431	/	HAS THE SPECIFIC RESPONSIBILITY TO ACCESS THE DATA IN A MANNER ANALOGOUS TO A	7
8	1432	/	TYPICAL DEVICE HANDLER, BUT ALSO HAS THE ADDITIONAL RESPONSIBILITY TO DEAL	8
9	1433	/	WITH PREVAILING WORDS KNOWN TO EXIST THAT ARE NOT SPECIFICALLY PART OF THE	9
10	1434	/	BINARY DATA. SINCE PDP-8 FORMAT DECTAPES ARE 129 WORDS PER BLOCK,	10
11	1435	/	DECTAPE-SPECIFIC SLURP LOADERS [FOR THE TC01/TC08 AND THE TD8E] MUST IGNORE	11
12	1436	/	THE THREE ADDITIONAL DATA WORDS IN THE BLOCK [AND ANY ADDITIONAL WORDS THAT	12
13	1437	/	ARE PRESENT AS PART OF THE OVERALL LOW-LEVEL DECTAPE FORMAT]. [NOTE: SLURP	13
14	1438	/	LOADERS FOR OTHER DEVICES GENERALLY FUNCTION IN AN ANALOGOUS MANNER; HOWEVER	14
15	1439	/	CERTAIN DEVICE-SPECIFIC QUIRKS MAY APPLY.] ALL P?S/8 RELEASES ALSO SUPPORT A	15
16	1440	/	DEVICE-INDEPENDENT VIRTUAL SLURP LOADER AS A COMMAND-LINE OPTION OF THE "BIN"	16
17	1441	/	SYSTEM PROGRAM THAT LAUNCHES A SLURP LOADER [WHEN LOADING THE CONTENTS OF	17
18	1442	/	BINARY FILES, ONE OF SEVERAL OPERATING MODES OF THE BIN PROGRAM]; FOR SYSTEM	18
19	1443	/	CONFIGURATIONS THAT DO NOT SUPPORT A DEVICE-SPECIFIC SLURP LOADER, THE VIRTUAL	19
20	1444	/	SLURP LOADER IS ALWAYS USED BY DEFAULT.	20
21	1445			21
22	1446	/	NOTE: SLURP BINARY FORMAT IS ALSO USED IN SEVERAL OTHER OPERATING SYSTEMS,	22
23	1447	/	INCLUDING CERTAIN OPERATING SYSTEMS SOLD BY DEC. [IT IS EXTREMELY LIKELY DEC	23
24	1448	/	MANAGEMENT PERSONNEL NEVER UNDERSTOOD THIS DESPITE THE OBVIOUS COPYRIGHT	24
25	1449	/	IMPLICATIONS.] THESE INCLUDE THE R-L MONITOR SYSTEM [WHICH WAS WRITTEN BY	25
26	1450	/	RICHARD LARY AND LENNY ELEKMAN WITH ADDITIONAL CONTRIBUTIONS BY HANK MAURER,	26
27	1451	/	JACK BURNES, JOE FISCHETTI, ROD DORMAN AND CHARLES LASNER], DIBOL-8 [A	27
28	1452	/	HACKED-UP VARIANT OF THE R-L MONITOR SYSTEM TRIVIAALLY CREATED BY HANK MAURER	28
29	1453	/	TO SUPPORT THE FIRST DIBOL COMPILER AS AN R-L MONITOR SYSTEM PROGRAM] AND ALL	29
30	1454	/	VARIANT RELEASES OF EDU-SYSTEM 30 [WHICH IS IN TURN BASED ON POLY BASIC, A	30
31	1455	/	PRODUCT OF RICHARD LARY AND LENNY ELEKMAN WHILE STUDENTS AT BROOKLYN POLYTECH	31
32	1456	/	AND AS SUCH, NOT EMPLOYEES OF DEC]. POLY BASIC WAS DEVELOPED AS A STANDALONE	32
33	1457	/	OPERATING SYSTEM ON THE R-L MONITOR, THE PROGENITOR SYSTEM FROM WHICH P?S/8 IS	33
34	1458	/	DESCENDED; WHILE THERE IS NO INTERNAL FILE SYSTEM ASSOCIATED WITH SLURP FORMAT	34
35	1459	/	IN THESE BASIC-ONLY SYSTEMS, IT IS THE INTERNAL FORMAT OF ALL BINARY	35
36	1460	/	OPERATIONS FOR THE INTERNAL RUN-TIME SYSTEM.	36
37	1461			37
38	1462	/	SLURP GROUP FORMAT.	38
39	1463			39
40	1464	/	EACH OF THE [UP TO] 18 GROUPS IN ANY ONE LOGICAL BLOCK IS PROCESSED IN	40
41	1465	/	SEQUENCE BEFORE MOVING ON TO THE NEXT BLOCK IN THE FILE [IF APPLICABLE]. THE	41
42	1466	/	CONTROL [OR FLAG] WORD [FOR THE NEXT SIX DATA WORDS] IS PROCESSED FIRST TO	42
43	1467	/	DETERMINE THE DISPOSITION OF EACH OF THE SIX DATA WORDS THAT FOLLOW.	43
44	1468			44
45	1469	/	EACH CONTROL WORD CONTAINS SIX PAIRS OF CONTROL BITS, EACH FOR THE	45
46	1470	/	CORRESPONDING DATA WORD. [BIT PAIRS ARE RECKONED LEFT TO RIGHT.]	46
47				47
48				48
49				49
50				50
51				51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1472	/	WITHIN EACH BIT PAIR, THE FOLLOWING DEFINITIONS APPLY:	1
2	1473			2
3	1474	/	00 - THE CORRESPONDING WORD IS DATA TO BE LOADED INTO THE LOCATION INDICATED	3
4	1475	/	BY THE CURRENT LOADING ORIGIN AND THE CURRENT LOADING FIELD. THE	4
5	1476	/	LOADING ORIGIN IS INCREMENTED FOR ANY ADDITIONAL DATA WORD LOADING.	5
6	1477			6
7	1478	/	10 - THE CORRESPONDING WORD IS AN ORIGIN SETTING; THE LOADING ORIGIN WILL BE	7
8	1479	/	UPDATED TO THE CONTENTS OF THIS WORD FOR ANY ADDITIONAL DATA WORD	8
9	1480	/	LOADING.	9
10	1481			10
11	1482	/	11 - THE CORRESPONDING WORD IS A FIELD SETTING EXPRESSED AS A "CDF X" [62X1]	11
12	1483	/	INSTRUCTION TO THE NEW LOADING FIELD X; THE CONTENTS OF THIS WORD WILL BE	12
13	1484	/	STORED INLINE TO CHANGE THE EFFECTIVE LOADING FIELD FOR ANY ADDITIONAL	13
14	1485	/	DATA WORDS.	14
15	1486			15
16	1487	/	01 - END OF FILE HAS BEEN DETECTED; IGNORE ALL ADDITIONAL DATA WORDS IN THIS	16
17	1488	/	DATA BLOCK AND ANY ADDITIONAL BLOCKS IN THE CURRENT FILE. THE SLURP	17
18	1489	/	LOADER WILL MOVE ON TO PROCESSING ADDITIONAL SLURP FILES [IF APPLICABLE].	18
19	1490			19
20	1491	/	NOTE: WHEN APPROPRIATE COMMAND-LINE OPTIONS ARE INVOKED, THE P?S/8 "BIN"	20
21	1492	/	PROGRAM IS ALSO CAPABLE OF READING BINARY PAPER-TAPES [FROM EITHER THE	21
22	1493	/	LOW-SPEED OR HIGH-SPEED READER] AND CONVERTING THE BINARY DATA INTO THE	22
23	1494	/	CONTENTS OF SPECIFIED SLURP BINARY TFS FILES. USING ADDITIONAL COMMAND-LINE	23
24	1495	/	OPTIONS, THE SLURP LOADER CAN BE INVOKED IN THE SAME COMMAND TO LOAD THE	24
25	1496	/	CONTENTS OF THE FILES CREATED EARLIER IN THE COMMAND OPERATION.	25
26	1497			26
27	1498	/	NOTE: THE P?S/8 "BIN" PROGRAM IS ALSO CAPABLE OF ADDITIONAL OPERATIONS TO	27
28	1499	/	PUNCH THE CONTENTS OF SLURP BINARY FILES OR TO PLACE A COPY OF THE RIM LOADER	28
29	1500	/	AND A CUSTOMIZED VERSION OF THE BIN LOADER INTO MEMORY WITH A COPY OF THE	29
30	1501	/	SLURP LOADER BOOTSTRAP AS DEFINED ABOVE. OTHER THAN THE DEVICE-DEPENDENT	30
31	1502	/	BOOTSTRAP WORDS, THIS OPERATION IS DEVICE-INDEPENDENT; APPROPRIATE	31
32	1503	/	COMMAND-LINE OPTIONS ARE AVAILABLE FOR ALL OPERATING MODES OF THE P?S/8 "BIN"	32
33	1504	/	SYSTEM PROGRAM.	33
34	1505			34
35	1506	/	SINCE NO CHECKSUM SCHEME IS USED IN THE SLURP FILE FORMAT, ADVANCED USERS CAN	35
36	1507	/	FREELY PATCH ANY BINARY FILE AS REQUIRED. THIS IS USEFUL WHEN MODIFYING	36
37	1508	/	PAPER-TAPE SOURCED/CONVERTED BINARY FILES WHICH MAY REQUIRE ALTERNATE MEMORY	37
38	1509	/	FIELDS THAN THE PROGRAM WAS ORIGINALLY ASSEMBLED FOR LOADING.	38
39	1510			39
40	1511	/	SELF-STARTING BINARY PROGRAM ISSUES.	40
41	1512			41
42	1513	/	"SPARM" [07756] CONTAINS THE 12-BIT STARTING ADDRESS OF THE PROGRAM. BY	42
43	1514	/	DIRECTLY LOADING INTO THIS ADDRESS, THE PROGRAM WILL ALWAYS START ACCORDING	43
44	1515	/	TO THE USER'S PREFERENCE.	44
45	1516			45
46	1517	/	"BCIF" [07746] CONTAINS THE STARTING FIELD IN THE FORM OF A "CIF CDF XX"	46
47	1518	/	INSTRUCTION WHERE XX IS THE STARTING FIELD [IN BITS(6-8)]. [NOTE: THIS	47
48	1519	/	SPECIFIC ADDRESS APPLIES TO VERSION 8Z AND IS SUBJECT TO CHANGE.]	48
49	1520			49
50	1521	/	NOTE: SELF-STARTING INFORMATION IS IGNORED IF THE VIRTUAL BINARY SLURP LOADER	50
51	1522	/	IS IN EFFECT [AS ALL ATTEMPTS TO LOAD INTO 07600-07777 ARE IGNORED].	51
52				52
53				53
54				54
55				55
56				56
57				57
58				58

1	1524	/	COMES HERE TO STOP THE LATEST DECTAPE DRIVE UNIT AT THE END OF EVERY FILE.	1
2	1525			2
3	1526	007605* 1350	BREAD2, TAD BUNIT /GET LATEST UNIT INTO AC[0].	3
4	1527	007606* 6774	SDLC /STOP THE LATEST TAPE UNIT WE WERE READING.	4
5	1528			5
6	1529	/	FALLS THROUGH TO PROCESS THE NEXT FILE [IF ANY].	6
7	1530			7
8	1531	/	THIS IS THE SLURP LOADER STARTING ADDRESS. THE FILE POINTER MUST BE	8
9	1532	/	INCREMENTED BEFORE USING TO OBTAIN THE LATEST FILE BLOCK AND UNIT BIT	9
10	1533	/	DATA. IF THE LATEST FILE LIST ELEMENT OBTAINED IS 0000 THERE ARE NO	10
11	1534	/	ADDITIONAL FILES TO LOAD. [NOTE: IF THE FILE LIST IS THE LARGEST SIZE ENDING	11
12	1535	/	IN 07777, THE 0000 VALUE IS OBTAINED BECAUSE THE "ISZ BFLPTR" INSTRUCTION	12
13	1536	/	SKIPS THE NEXT INSTRUCTION. AS SUCH, ONLY SHORTER FILE LISTS ARE TERMINATED	13
14	1537	/	WITH A SENTINEL 0000 VALUE PAST THE LAST PASSED FILE ELEMENT.]	14
15	1538			15
16	1539	007607* 2354	BREAD, ISZ BFLPTR /BUMP TO NEXT FILE.	16
17	1540	007610* 1754	TAD I BFLPTR /GET THE NEXT FILE HANDLE WORD.	17
18	1541	007611* 0353	AND B7770/(7770) /JUST FILE BLOCK BITS.	18
19	1542	007612* 7450	SNA /SKIP IF AN ACTVIE ENTRY.	19
20	1543	007613* 5334	JMP BCIF /JUMP IF ALL FILES PROCESSED.	20
21	1544	007614* 3347	DCA BBLK /SAVE LATEST FILE BLOCK.	21
22	1545	007615* 1754	TAD I BFLPTR /GET FILE HANDLE AGAIN.	22
23	1546	007616* 7010	RAR /LOW-ORDER UNIT BIT TO LINK.	23
24	1547	007617* 7210	CLA RAR /LOW-ORDER UNIT BIT TO AC[0].	24
25	1548	007620* 3350	DCA BUNIT /SAVE UNIT BIT FOR NOW.	25
26	1549	007621* 7710	BCONT, SPA CLA /SKIP NEXT IF CLEAR.	26
27	1550	007622* 7330	BTURNX, NL4000 /SET REVERSING BIT.	27
28	1551	007623* 1351	TAD BDIRGO /GET CURRENT DIRECTION AND GO BITS.	28
29	1552	007624* 3351	DCA BDIRGO /SAVE [WITH POTENTIALLY UPDATED] DIRECTION.	29
30	1553	007625* 1351	TAD BDIRGO /GET CURRENT DIRECTION AND GO BITS	30
31	1554	007626* 7110	CLL RAR /MOVE TO HARDWARE DIRECTION AND GO BITS.	31
32	1555	007627* 1350	TAD BUNIT /ADD ON CURRENT UNIT BIT.	32
33	1556	007630* 6774	SDLC /GET THE TAPE MOVING IN THE DESIRED DIRECTION.	33
34	1557	007631* 4327	JMS BGETWD /WAIT FOR	34
35	1558	007632* 4327	JMS BGETWD /SOME LINES.	35
36	1559	007633* 6771	BSRCHMO, SDSS /SINGLE LINE FLAG UP?	36
37	1560	007634* 5233	JMP .-1 /NO, WAIT FOR IT.	37
38	1561	007635* 6776	SDRC /YES, READ THE COMMAND REGISTER.	38
39	1562	007636* 7106	CLL RTL /MOVE UP TWO BITS; DIRECTION TO LINK.	39
40	1563	007637* 0266	AND B374/(374) /JUST MARK TRACK BITS.	40
41	1564	007640* 1201	TAD B7670/(-110) /COMPARE TO END ZONE VALUE.	41
42	1565	007641* 7450	SNA /SKIP IF NOT IN THE END ZONE.	42
43	1566	007642* 5222	JMP BTURNX /REVERSE DIRECTION IF SO.	43
44	1567	007643* 1346	TAD B7760/(-130+110) /COMPARE TO BLOCK MARK VALUE.	44
45	1568	007644* 7640	SZA CLA /SKIP IF AT BLOCK MARK.	45
46	1569	007645* 5233	JMP BSRCHMORE /GO BACK AND KEEP SEARCHING.	46
47	1570	007646* 6777	SDRD /GET LATEST BLOCK.	47
48	1571	007647* 7430	SZL /SKIP IF GOING FORWARD.	48
49	1572	007650* 1352	TAD B4002 /ADD ON UNDERSHOOT VALUE IF REVERSE SEARCH.	49
50	1573	007651* 7041	CIA /INVERT FOR TESTING.	50
51	1574	007652* 1347	TAD BBLK /COMPARE TO DESIRED BLOCK.	51
52	1575	007653* 7440	SZA /SKIP IF IT MATCHES.	52
53	1576	007654* 5221	JMP BCONT /DETERMINE WHAT TO DO NEXT THERE.	53
54				54
55				55
56				56
57				57
58				58

1	1578	/	THE SELECTED DECTAPE IS MOVING FORWARD AND THE DESIRED BLOCK MARK FOUND.			1
2	1579					2
3	1580	/	NOW WE NEED TO LOCATE THE 40 MARK WINDOW VALUE [PART OF THE LOCK WORD].			3
4	1581					4
5	1582	007655* 6771	BFNDLOC, SDSS		/SINGLE LINE FLAG UP?	5
6	1583	007656* 5255	JMP	.-1	/NO, WAIT FOR IT.	6
7	1584	007657* 6776	SDRC		/YES, GET THE COMMAND REGISTER.	7
8	1585	007660* 0343	AND	B77/(77)	/JUST MARK WINDOW BITS.	8
9	1586	007661* 1265	TAD	B7740/(-40)	/COMPARE TO MARK WINDOW VALUE [LOCK MARK].	9
10	1587	007662* 7640	SZA CLA		/SKIP IF IT MATCHES.	10
11	1588	007663* 5255	JMP	BFNDLOCK40	/JUMP IF NOT.	11
12	1589					12
13	1590	/	READ ONE MORE WORD; WE WILL BE LOCATED AT THE FIRST DATA WORD OF THE BLOCK.			13
14	1591					14
15	1592	007664* 4327	JMS	BGETWD	/READ THE LAST NON-DATA WORD.	15
16	1593	007665* 7740	B7740, SMA SZA CLA		/CLEAN UP; THIS MIGHT SKIP.	16
17	1594	007666* 0374	B374, 374		/CONSTANT 0374; EXECUTES AS A HARMLESS	17
18	1595		/"AND" INSTRUCTION IF NOT SKIPPED.			18
19	1596					19
20	1597	/	START LOADING THE LATEST BLOCK BINARY DATA.			20
21	1598					21
22	1599	007667* 1345	BNWREC, TAD	B7756/(-22)	/22 GROUPS IN ONE BLOCK.	22
23	1600	007670* 3340	DCA	BGRPCT	/INTO GROUP COUNTER.	23
24	1601	007671* 1342	BNWGRP, TAD	B7772/(-6)	/6 BINARY WORDS IN A GROUP.	24
25	1602	007672* 3341	DCA	BWDCT	/INTO WORD COUNTER.	25
26	1603	007673* 4327	JMS	BGETWD	/GET LATEST CONTROL WORD.	26
27	1604	007674* 7110	CLL RAR		/DO AN INITIAL SHIFT.	27
28	1605	007675* 7410	SKP		/DON'T GET RUNNING VALUE FIRST TIME THROUGH.	28
29	1606	007676* 1337	BGRPLP, TAD	BRLBTS	/GET CURRENT CONTROL WORD.	29
30	1607	007677* 7006	RTL		/ROTATE NEXT PAIR TO LINK AND AC[0].	30
31	1608	007700* 3337	DCA	BRLBTS	/STORE BACK FOR NEXT TIME.	31
32	1609	007701* 1337	TAD	BRLBTS	/RESTORE LATEST TO AC[0].	32
33	1610	007702* 7710	SPA CLA		/<EOF> OR FIELD SETTING?	33
34	1611	007703* 5322	JMP	BE0F	/YES. [LINK IS OFF IF <EOF>]	34
35	1612	007704* 4327	JMS	BGETWD	/NO, GET DATA OR ORIGIN VALUE.	35
36	1613	007705* 6201	BCDF, CDF	00+.-.	/PREVAILING FIELD BITS WILL BE FILLED IN.	36
37	1614	007706* 7420	SNL		/SKIP IF ORIGIN SETTING.	37
38	1615	007707* 3736	DCA I	BLOC	/STORE DATA WORD.	38
39	1616	007710* 6201	CDF	00	/RESTORE OUR FIELD.	39
40	1617	007711* 2336	ISZ	BLOC	/BUMP STORAGE POINTER [EVEN IF WRONG TO DO SO].	40
41	1618	007712* 7430	SZL		/SKIP IF DATA WORD; UNIMPORTANT IF SKIPPED.	41
42	1619	007713* 3336	DCA	BLOC	/SET NEW ORIGIN IF NOT DATA WORD.	42
43	1620	007714* 2341	BNXTWD, ISZ	BWDCT	/DONE SIX WORDS IN THIS GROUP YET?	43
44	1621	007715* 5276	JMP	BGRPLP	/NO, GO DO ANOTHER ONE	44
45	1622	007716* 2340	ISZ	BGRPCT	/DONE 22 GROUPS IN THIS BLOCK YET?	45
46	1623	007717* 5271	JMP	BNWGRP	/NO, GO DO ANOTHER ONE.	46
47	1624					47
48	1625	/	ALL BINARY DATA IN THIS BLOCK IS NOW LOADED; GO ON TO THE NEXT BLOCK.			48
49	1626					49
50	1627	007720* 2347	ISZ	BBLK	/BUMP TO NEXT BLOCK.	50
51	1628	007721* 5221	JMP	BCONT	/KEEP GOING.	51
52						52
53						53
54						54
55						55
56						56
57						57
58						58

1	1630	/	COMES HERE IF THE LATEST WORD IS EITHER A FIELD SETTING OR AT THE END OF DATA	1
2	1631	/	WITHIN THE LATEST FILE.	2
3	1632			3
4	1633	007722* 7620	BE0F, SNL CLA /FIELD SETTING?	4
5	1634	007723* 5205	JMP BREAD2 /NO, IT'S THE END OF THIS FILE.	5
6	1635	007724* 4327	JMS BGETWD /YES, GET THE NEXT WORD.	6
7	1636	007725* 3305	DCA BCDF /STORE IN-LINE AS A "CDF" INSTRUCTION	7
8	1637		/TO THE LATEST FIELD.	8
9	1638	007726* 5314	JMP BNXTWD /GO DO ANOTHER WORD.	9
10	1639			10
11	1640	007727* 0000	BGETWD, ... /GET A NEW WORD ROUTINE.	11
12	1641	007730* 6773	SDSQ /QUAD LINE FLAG UP?	12
13	1642	007731* 5330	JMP ...-1 /NO, WAIT FOR IT.	13
14	1643	007732* 6777	SDRD /YES, GET THE DATA WORD.	14
15	1644	007733* 5727	JMP I BGETWD /RETURN TO CALLER.	15
16	1645			16
17	1646	/	WHEN ALL FILES HAVE BEEN LOADED, START THE PROGRAM HERE. THE "BIN" PROGRAM	17
18	1647	/	STORED THE STARTING FIELD HERE; THE STARTING ADDRESS IS IN "SPARM" [07756].	18
19	1648			19
20	1649	007734* 6203	BCIF, ...^10+CIF CDF /CHANGE FIELD TO STARTUP FIELD	20
21	1650	007735* 5756	JMP I SPARM /AND START THE USER PROGRAM.	21
22	1651			22
23	1652	007736* 0000	BLOC, ... /MEMORY STORAGE POINTER.	23
24	1653	007737* 0000	BRLBTS, ... /CONTROL FLAGS WORD STORED HERE.	24
25	1654	007740* 0000	BGRPCTR, ... /GROUP COUNTER.	25
26	1655	007741* 0000	BWDCT, ... /WORD COUNTER.	26
27	1656			27
28	1657	/	DEFAULT STARTING ADDRESS [SAFE HALT].	28
29	1658			29
30	1659	BERR,	/DEFAULT STARTING ADDRESS.	30
31	1660	007742* 7772	B7772, SPA SNA SZL CLA HLT /THIS HALTS.	31
32	1661	007743* 0077	B77, 77 /CONSTANT 0077; IF CONTINUE IS PRESSED, THIS	32
33	1662		/EXECUTES AS A HARMLESS "AND" INSTRUCTION THAT	33
34	1663		/CAN BE SKIPPED.	34
35	1664			35
36	1665	/	IN CASE CONTINUE IS PRESSED, JMP BACK TO THE SAFE HALT.	36
37	1666			37
38	1667	007744* 5342	JMP BERR /INFINITE LOOP IF CONTINUE PRESSED.	38
39	1668			39
40	1669	007745* 7756	B7756, 7756 /CONSTANT 7756.	40
41	1670	007746* 7760	B7760, 7760 /CONSTANT 7760.	41
42	1671	007747* 0000	BBLK, ... /LATEST FILE BLOCK.	42
43	1672	007750* 0000	BUNIT, ... /CURRENT DRIVE IN BIT[0].	43
44	1673	007751* 2000	BDIRGO, 2000+... /CURRENT TAPE DIRECTION AND GO BITS.	44
45	1674	007752* 4002	B4002, 4002 /CONSTANT 4002.	45
46	1675	007753* 7770	B7770, 7770 /CONSTANT 7770.	46
47	1676	007754* 7756	BFLPTR, SFILES-1 /FILE LIST POINTER [INCREMENT BEFORE USING].	47
48	1677	007755* 0000	0	48
49	1678	007756* 7742	SPARM, BERR /STARTING ADDRESS PASSED HERE [DEFAULT SET].	49
50	1679	007757* 0000	SFILES, ZBLOCK ... /FILE HANDLES PASSED HERE.	50
51	1680			51
52	1681	*4600	RELOC /TURN OFF RELOCATION.	52
53				53
54				54
55				55
56				56
57				57
58				58

1	1683	/	P?S/8 SYSTEM DEVICE HANDLER RELOADER.	1
2	1684			2
3	1685	/	WHEN THE DEVICE-DEPENDENT BINARY FILE SLURP LOADER [AS OPPOSED TO THE	3
4	1686	/	DEVICE-INDEPENDENT VIRTUAL BINARY FILE SLURP LOADER SUPPORTED ON ALL P?S/8	4
5	1687	/	SYSTEMS] IS IN EFFECT, THE CONTENTS OF 07600-07777 NO LONGER CONTAIN THE	5
6	1688	/	SYSTEM DEVICE HANDLER; A REBOOT ROUTINE IS PROVIDED AT 07600 [AND SOME	6
7	1689	/	ADDITIONAL WORDS FOLLOWING] TO ALLOW A TOTAL SYSTEM RELOAD CONSISTENT WITH	7
8	1690	/	THE MORE USUAL RESTARTING OF THE SYSTEM AT THE SAME ADDRESS.	8
9	1691			9
10	1692	/	AT FIRST GLANCE, IT WOULD APPEAR THAT AS PART OF THE TRADEOFF OF LOADER	10
11	1693	/	CHOICE, EITHER MORE EFFICIENT BINARY FILE LOADING [WITH NO NEED TO	11
12	1694	/	WRITE-ENABLE THE SYSTEM DEVICE] AND THE LACK OF ACCESS TO THE NORMAL SYSTEM	12
13	1695	/	KERNEL IS PROVIDED [OR THE SYSTEM DEVICE MUST BE WRITE-ENABLED AND THE SYSTEM	13
14	1696	/	KERNEL IS RESIDENT IN MEMORY AS IS NORMALLY THE CASE].	14
15	1697			15
16	1698	/	HOWEVER, THIS IS NOT THE CASE. BY INVOKING THE P?S/8 "BIN" PROGRAM FOR BINARY	16
17	1699	/	FILE LOADING AND INCLUDING THE '/I' COMMAND-LINE OPTION SWITCH, AT THE END OF	17
18	1700	/	ALL LOADING OF THE ENTIRETY OF THE CONTENTS OF ALL BINARY FILES, IT IS	18
19	1701	/	POSSIBLE TO ALSO HAVE THE SYSTEM DEVICE HANDLER AS WELL! THERE ARE A FEW	19
20	1702	/	PROVISOS:	20
21	1703			21
22	1704	/	1) THE BLOCK JUST PAST THE END OF THE LAST FILE LOADED FROM THE SYSTEM	22
23	1705	/	DEVICE WILL DETERMINE THE FINAL POSITION OF THE DEVICE. [NOTE:	23
24	1706	/	THE DESCRIPTION USED HERE ENVISIONS DECTAPE AND RELATED SYSTEMS].	24
25	1707	/	HOWEVER, IF THE '/I' COMMAND-LINE OPTION SWITCH IS INVOKED, THERE WILL BE	25
26	1708	/	SEVERAL ADDITIONAL DEVICE OPERATIONS THAT WILL TEND TO LEAVE THE	26
27	1709	/	SYSTEM DEVICE POSITIONED JUST PAST BLOCK 0000 [WHICH WILL BE RELOADED	27
28	1710	/	AS PART OF THE EXTRA TAPE MOTION].	28
29	1711			29
30	1712	/	ASSUMING A PROGRAM NORMALLY [EVENTUALLY] EXITS TO 07600, THIS IS NOT	30
31	1713	/	AN OVERALL ISSUE PER SE; THE DEVICE MUST BE REWOUND TO BLOCK 0000 AS	31
32	1714	/	PART OF THE SYSTEM RESTART OPERATION. AS SUCH, THIS IS MERELY A	32
33	1715	/	MATTER OF REWINDING BEFORE THE PROGRAM STARTS AS OPPOSED TO WHEN THE	33
34	1716	/	PROGRAM EXITS.	34
35	1717			35
36	1718	/	2) PART OF THE ADDITIONAL OPERATIONS IS A BINARY LOADING OPERATION OF	36
37	1719	/	BLOCK 0060 ON THE SYSTEM DEVICE. THIS IS THE MECHANISM THAT PERFORMS	37
38	1720	/	THE SEEMINGLY IMPOSSIBLE TASK OF CONVERTING THE CONTENTS OF MEMORY	38
39	1721	/	LOCATIONS 07600-07777 FROM THE SYSTEM RESTART BOOTSTAP [AND BINARY	39
40	1722	/	LOADER ARTIFACTS] INTO THE CONTENTS OF BLOCK 0000 [THE IMAGE OF THE	40
41	1723	/	SYSTEM KERNEL].	41
42	1724			42
43	1725	/	3) BLOCK 0060 IS APPENDED TO THE LIST OF SLURP BINARY INPUT FILES; THE	43
44	1726	/	RELOAD OPERATION MUST BE PERFORMED AFTER THE NORMAL BINARY LOADING IS	44
45	1727	/	OTHERWISE COMPLETE. AS SUCH, BLOCK 0060 BECOMES THE LAST FILE HANDLE	45
46	1728	/	IN THE PASSED FILE LIST. [THE CONTENTS OF BLOCK 0060 ARE GENERALLY	46
47	1729	/	CONSISTENT WITH A SMALL SLURP BINARY FILE THAT ENDS WITHIN THE FIRST	47
48	1730	/	BLOCK.]	48
49	1731			49
50	1732	/	IF THE '/I' COMMAND-LINE OPTION IS IN EFFECT, THE MAXIMUM NUMBER OF	50
51	1733	/	INPUT FILES IS LOWERED FROM 17 TO 16 TO ENSURE A FILE HANDLE CAN BE	51
52	1734	/	PLACED AT THE END OF THE FILE LIST.	52
53				53
54				54
55				55
56				56
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1	1736	/			SINCE SLURP BINARY FILE FORMAT APPROACHES 6/7 OF THE LOADING	1
2	1737	/			EFFICIENCY OF CORE-IMAGE DATA, THIS LOWERS THE MAXIMUM AMOUNT OF	2
3	1738	/			BINARY DATA THAT CAN BE LOADED WITH A SINGLE COMMAND LINE FROM	3
4	1739	/			APPROXIMATELY 7.25 MEMORY FIELDS TO 6.9 MEMORY FIELDS; HOWEVER, SINCE	4
5	1740	/			TYPICAL PROGRAM LOADING FOR PRACTICAL PROGRAMMING PROJECTS SELDOM	5
6	1741	/			EXCEEDS 8K OF PROGRAM MEMORY, THIS IS NOT AN IMPORTANT CONSIDERATION.	6
7	1742	/			[NOTE: EXTREMELY LARGE PROGRAMS MIGHT REQUIRE TWO COMMANDS TO LOAD	7
8	1743	/			ALL PROGRAM COMPONENTS.]	8
9	1744	/				9
10	1745	/			FUTURE RELEASES OF P?S/8 WILL INCLUDE A BINARY CONVERSION PROGRAM	10
11	1746	/			KNOWN AS "BINCON" WHICH HAS SEVERAL MODES OF OPERATION. ADVANCED	11
12	1747	/			USERS OF BINCON WILL BE ABLE TO CONCATENATE THE CONTENTS OF A GROUP OF	12
13	1748	/			TFS SLURP BINARY FILES INTO A SINGLE EXTENDED-LENGTH FILE; AS SUCH,	13
14	1749	/			PROGRAM LOADING WILL BE MORE EASILY ACCOMPLISHED IN FEWER UNIQUE FILE	14
15	1750	/			NAMES. THE MINOR RESTRICTION ON MAXIMUM PASSED FILES WILL BECOME	15
16	1751	/			UNIMPORTANT.	16
17	1752	/				17
18	1753	/			THE FOLLOWING PROGRAMMING GENERATES THE IMAGE IN MEMORY OF A CONTRIVED SLURP	18
19	1754	/			BINARY FILE TO BE WRITTEN OUT TO BLOCK 0060 OF THE SYSTEM DEVICE. [NOTE:	19
20	1755	/			SINCE THIS BLOCK IS READ ONLY BY THE SLURP LOADER, IT DOES NOT HAVE TO CONFORM	20
21	1756	/			TO THE USUAL ORIGIN AND FIELD SETTING CONVENTIONS; THIS IS DONE TO MINIMIZE	21
22	1757	/			THE SIZE OF THE DATA, ALL OF WHICH MUST FIT INTO A SINGLE BLOCK.]	22
23	1758					23
24	1759	0002	FIELD	IGEN&70%10	/RELOADER BLOCK GENERATION FIELD.	24
25	1760	*4000		*IGEN&7600	/RELOADER BLOCK GENERATION ADDRESS.	25
26	1761					26
27	1762					27
28	1763		ICOR,	NOPUNCH	/FOOL THE ASSEMBLER.	28
29	1764					29
30	1765	0000	FIELD	0	/WHERE THIS LOADS.	30
31	1766					31
32	1767	*4000		*ICOR	/RESTORE ORIGIN [IN CASE FIELD CHANGES IT].	32
33	1768					33
34	1769			ENPUNCH	/UN-FOOL THE ASSEMBLER.	34
35	1770					35
36	1771	004000	5640	OR^4+FL^4+OR^4+OR^4+DA^4+DA	/CONTROL WORD.	36
37	1772					37
38	1773	004001	0200	0200	/*200. NOTE: THIS IS SUPERFLUOUS.	38
39	1774					39
40	1775	004002	6201	CDF 00	/FIELD 0.	40
41	1776					41
42	1777	004003	0200	0200	/*200. NOTE: THIS IS SUPERFLUOUS.	42
43						43
44						44
45						45
46						46
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1	1779	/	MOST OF THE CODE LOADS IN OVER THE [NOW UNNEEDED] FILE LIST AREA.		1
2	1780				2
3	1781	004004 7757	SFILES	/*SFILES; THIS IS WHERE THE CODE STARTS LOADING.	3
4	1782				4
5	1783	*7757	RELOC SFILES	/FOOL THE ASSEMBLER.	5
6	1784				6
7	1785	/	SINCE THE RELOADER SLURP FILE BLOCK IS THE LAST ELEMENT IN THE FILE LIST, THE		7
8	1786	/	ENTIRE FILE LIST AREA IS AVAILABLE FOR THE PURPOSE OF [SELECTIVELY] USING THE		8
9	1787	/	SLURP LOADER CODE FOR THE PURPOSE OF RELOADING THE SYSTEM DEVICE HANDLER. THE		9
10	1788	/	FOLLOWING CODE LOADS IN THE FILE LIST AREA WHICH IS ALLOWED AT THIS POINT.		10
11	1789				11
12	1790	007757* 7600	RLPOINT, SBOOT/CLA!400	/CLEAR THE AC; ALSO STORAGE POINTER.	12
13	1791				13
14	1792	007760* 1334	TAD BCIF	/GET PROGRAM STARTING FIELD INSTRUCTION.	14
15	1793				15
16	1794	007761* 0000	DA^4+DA^4+DA^4+DA^4+DA^4+DA	/NEXT CONTROL WORD.	16
17	1795				17
18	1796	*7761	RELOC .-1	/FOOL THE ASSEMBLER.	18
19	1797				19
20	1798	007761* 3372	DCA 7772	/STORE OUT OF HARM'S WAY.	20
21	1799				21
22	1800	/	NOW READ THE SYSTEM DEVICE HANDLER IN ONE WORD AT A TIME.		22
23	1801				23
24	1802	007762* 6773	RELOOP, SDSQ	/QUAD LINE FLAG UP?	24
25	1803	007763* 5362	JMP .-1	/NO, WAIT FOR IT.	25
26	1804	007764* 6777	SDRD	/YES, GET THE DATA WORD.	26
27	1805	007765* 3757	DCA I RLPOINTER	/STORE A WORD.	27
28	1806	007766* 2357	ISZ RLPOINTER	/BUMP TO NEXT.	28
29	1807				29
30	1808	007767* 0000	DA^4+DA^4+DA^4+DA^4+DA^4+DA	/NEXT CONTROL WORD.	30
31	1809				31
32	1810	*7767	RELOC .-1	/FOOL THE ASSEMBLER.	32
33	1811				33
34	1812	007767* 2374	ISZ RLCNTR	/DONE ENOUGH WORDS?	34
35	1813	007770* 5362	JMP RELOOP	/NO, KEEP GOING.	35
36	1814	007771* 6774	SDLC	/YES, STOP THE DECTAPE.	36
37	1815	007772* 6203	RLCIF, .-.^10+CIF CDF	/WILL BE THE STARTING FIELD INSTRUCTION.	37
38	1816	007773* 5756	JMP I SPARM	/GO START THE PROGRAM UP.	38
39	1817				39
40	1818	007774* 7622	RLCNTR, SBOOT-SPARM	/COUNT OF WORDS TO MOVE.	40
41					41
42					42
43					43
44					44
45					45
46					46
47					47
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1	1820	/		THE RELOADER CODE HAS NOW BEEN READ INTO THE FILE LIST AREA; THE SLURP LOADER	1
2	1821	/		WILL NOW BE MODIFIED TO READ IN THE CONTENTS OF BLOCK 0000.	2
3	1822				3
4	1823	007775*	4210	OR^4+DA^4+OR^4+DA^4+OR^4+DA /NEXT CONTROL WORD.	4
5	1824				5
6	1825	/		ALL OF THE RELOADER CODE WORDS HAVE BEEN READ IN. THE SLURP LOADER IS NO	6
7	1826	/		LONGER NEEDED FOR ITS NORMAL PURPOSE. THE SLURP LOADER WILL NOW BE MODIFIED	7
8	1827	/		TO CAUSE IT TO READ IN BLOCK 0000 DATA.	8
9	1828				9
10	1829	/		AFTER THE END OF FILE IS DETECTED IN THE [STILL-FUNCTIONING] SLURP LOADER, THE	10
11	1830	/		CODE GOES BACK TO [NORMALLY] DETECT THE END OF THE FILE LIST [WHICH WOULD THEN	11
12	1831	/		START THE PROGRAM UP]. BEFORE THAT HAPPENS, IT WILL BE MODIFIED TO INSTEAD	12
13	1832	/		SEARCH FOR BLOCK 0000. WHEN THE SEARCH COMPLETES AND BLOCK 0000 DATA IS	13
14	1833	/		AVAILABLE, PREVENT THE SLURP LOADER READ OPERATIONS, AND INSTEAD EXECUTE THE	14
15	1834	/		RELOAD CODE READ IN OVER THE [AVAILABLE] FILE LIST AREA.	15
16	1835				16
17	1836	*7667		RELOC BNWREC /FOOL THE ASSEMBLER.	17
18	1837				18
19	1838	007667*	7667	BNWREC /*BNWREC.	19
20	1839				20
21	1840	007670*	5357	JMP RLPOINTER /RELOAD BLOCK 0000 INSTEAD OF SLURP LOADING.	21
22	1841				22
23	1842	/		THE SLURP LOADER SEARCH ROUTINE SUPPORTS BOTH DRIVE UNITS. THE FOLLOWING CODE	23
24	1843	/		FORCES ACCESS TO THE SYSTEM DRIVE UNIT [0].	24
25	1844				25
26	1845	*7627		RELOC BTURNX+5 /FOOL THE ASSEMBLER.	26
27	1846				27
28	1847	007627*	7627	BTURNX+5 /*BTURNX+5.	28
29	1848				29
30	1849	007630*	7000	NOP /THIS FORCES THE SYSTEM DEVICE UNIT [0].	30
31	1850				31
32	1851	/		WHEN THE SLURP LOADER GOES BACK TO DETECT ANOTHER FILE, FORCE IT TO INSTEAD	32
33	1852	/		USE BLOCK 0000 AND START A SEARCH ON THAT BLOCK.	33
34	1853				34
35	1854	*7607		RELOC BREAD /FOOL THE ASSEMBLER	35
36	1855				36
37	1856	007607*	7607	BREAD /*BREAD.	37
38	1857				38
39	1858	007610*	5214	JMP BREAD+5 /FORCE BLOCK 0000 TO BE LOCATED FOR READING.	39
40	1859				40
41	1860	*7771		RELOC SFILES+12 /FOOL THE ASSEMBLER [REALLY DOESN'T MATTER].	41
42	1861				42
43	1862	007771*	2525	EF^4+EF^4+EF^4+EF^4+EF^4+EF /NEXT CONTROL WORD.	43
44	1863				44
45	1864	007772*	0000	ZBLOCK 6 /NO DATA, JUST END OF FILE.	45
46	1865				46
47	1866	/		THIS IS THE END OF FILE. THE SLURP LOADER WILL NOW GO BACK AND START LOADING	47
48	1867	/		IN [MOST OF] BLOCK 0000.	48
49					49
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1	1869	*4043	RELOC	/TURN OFF RELOCATION.	1
2	1870				2
3	1871	/	END OF THE RELOADER FILE IMAGE WHICH MUST FIT IN ONE PAGE. CHECK TO MAKE SURE		3
4	1872	/	THE CODE DOES NOT EXCEED THIS RESTRICTION. [MOST SLURP FILES ARE SEVERAL		4
5	1873	/	BLOCKS LONG; THIS IS NOT CONSIDERED A "FILE".]		5
6	1874				6
7	1875	4200	ICOREND=ICOR+200	/FIRST LOCATION AFTER MAXIMUM	7
8	1876			/SIZE RELOADER CODE IMAGE.	8
9	1877				9
10	1878	4043	IMAGEND=.	/FIRST LOCATION AFTER ACTUAL RELOADER IMAGE.	10
11	1879				11
12	1880	0135	IM=ICOREND-IMAGEND	/COUNT OF ADDITIONAL WORDS AVAILABLE.	12
13	1881				13
14	1882		IFNZRO IM&4000 <ERROR .>	/ENSURE CODE LIMIT NOT EXCEEDED.	14
15					15
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1	1885	/	STUFF TO PUT IN SOMEWHERE ELSE LATER. [AFTER I FIGURE OUT MORE THINGS!]	1
2	1886			2
3	1887	/	SYSTEM NAME TDRM [THIS IS MEANINGFUL TO THE LOGICAL CONSOLE OVERLAY SUPPORT	3
4	1888	/	SECTION OF THIS FILE].	4
5	1889			5
6	1890	/	FOR NOW, THE MR8E-C SOURCE CODE IS NOT INCLUDED; IT IS NEEDED DURING THE	6
7	1891	/	ACTUAL USAGE ON NON-ROM SYSTEMS. HOWEVER, THE SOURCE CODE IN THIS SECTION	7
8	1892	/	CAN BE ASSEMBLED WITHOUT THE INCLUSION OF THAT SOURCE MODULE [BEING DEVELOPED	8
9	1893	/	SEPARATELY AND IN PRE-FINAL STATE]. WHEN IT IS STABLE, IT WILL BE ADDED TO	9
10	1894	/	THIS SOURCE FILE FOR COMPLETENESS.	10
11	1895			11
12	1896	\$	/THAT'S ALL FOR NOW, FOLKS!	12
13				13
14				14
15				15
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1					1
2					2
3	ADRLST=7603	DSPBLB=0024	READLP=7677	SYSBLK=0000	3
4	AT7577 7577	EF =0001	RELOOP 7762	SYSIO 7640	4
5	BLEN =0043	ENDWAI 1564	REWIND 1547	SYSNUM=0014	5
6	BBLK 7747	EQUFUN 7704	RLCIF 7772	SYSTAR 7637	6
7	BCDF 7705	EQUIMP 7750	RLCNTR 7774	S5671 7776	7
8	BCIF 7734	ERRTRP=0046	RLPOIN 7757	S6000 7642	8
9	BCONT 7621	ESTATU=0077	ROMBOO=7470	S70 7727	9
10	BDIRGO 7751	FILBLB=0023	ROMFLD=0070	S7405 7775	10
11	BEOF 7722	FL =0003	ROMGO =7405	S7600 7772	11
12	BERR 7742	F2SET =7552	SBLK 7734	TADGTC 7553	12
13	BFLPTR 7754	GENAD1=7400	SBLOCK 7635	TAPMOV 1556	13
14	BFNDLO 7655	GET 7312	SBOOT 7600	TDSIZE=2702	14
15	BGETWD 7727	GETFLD 7650	SBTFUN 7602	TEMP1 =0100	15
16	BGRPCT 7740	GETWRD 7647	SCIDF 7746	TRAPIO=1041	16
17	BGRPLP 7676	GTCMPV 7575	SCKSUM 7733	TRPERR=1200	17
18	BINADR=0020	GTCT 7561	SCORE 7633	TRPGEN=1020	18
19	BINIT =0020	GTLOOP 7565	SCRSIZ 7611	TRYCNT=7730	19
20	BINITL=0020	GTMARK 7544	SDATE 7610	UNIT =0000	20
21	BINIT2=0031	GTSRCH 7547	SDCODE=0077	WAITIO=1110	21
22	BISTR7=7776	GT1000 7545	SDLC =6774	WCT 7326	22
23	BLKDAT=7360	GT77 7560	SDLD =6775	WORDCT 7754	23
24	BLOC 7736	HIDWRD=7600	SDRC =6776	WRITE =4000	24
25	BNWGRP 7671	ICOR 4000	SDRD =6777	WRQ =7724	25
26	BNWREC 7667	ICOREN=4200	SDSQ =6773	WRQUAD 7671	26
27	BNXTWD 7714	IGEN =4020	SDSS =6771	ZAPLST=7627	27
28	BOOTBU=7354	IM =0135	SDST =6772		28
29	BREAD 7607	IMAGEN=4043	SERR 7661		29
30	BREAD2 7605	INSERT=1305	SEXIT 7722		30
31	BRLBTS 7737	IOTRAP=0045	SFILES 7757		31
32	BROMBO 7602	IOWAIT=0047	SFLDXI 7725		32
33	BSRCHM 7633	K1000 7300	SFROM 7773		33
34	BTURNX 7622	K77 7331	SFUN 7634		34
35	BUFADR 7751	L200 =0105	SFUNCT 7732		35
36	BUFPR 7753	L3000 1546	SF2SET 7730		36
37	BUNIT 7750	L7 =0177	SINTXI 7724		37
38	BUTSTR 7562	L7000 =0077	SLBTEN=7605		38
39	BWDCT 7741	L77 =0056	SLOAD 7632		39
40	B374 7666	L7756 =0064	SLPCOR 4400		40
41	B4002 7752	L7770 =0040	SLPGEN=4420		41
42	B7670 7601	MARKSE 7321	SLPSA =7777		42
43	B77 7743	MONSTR=0353	SLUPAG=2600		43
44	B7740 7665	MOVBLB=0022	SMORE 7612		44
45	B7756 7745	MRKSR 7315	SMORE2 7736		45
46	B7760 7746	MRKVAL 7327	SMORE3 7665		46
47	B7770 7753	NL0001=7201	SMVLUP 7764		47
48	B7772 7742	NL2000=7332	SOUTFL 7607		48
49	CALADR=0020	NL4000=7330	SPARM 7756		49
50	CALFLD=0021	NL7776=7344	SROMGO 7670		50
51	CHLOOP=1416	OR =0002	STARTI=7400		51
52	CIFR 7674	PAGCNT 7752	STO 7774		52
53	CNTLST=7615	PCHLOO=0162	STRT 7332		53
54	DA =0000	PUTFLD 7655	SUNIT 7735		54
55	DESELE=1573	PUTWRD 7654	SWAL 7604		55
56	DEVSIZ=6760	RD 7303	SWMX 7605		56
57	DRVUNI 7731	RDQUAD 7676	SWY9 7606		57
58					58

1		1
2		2
3	NO ERRORS DETECTED	3
4		4
5	NO LINKS GENERATED	5
6		6
7	6K MEMORY UTILIZED	7
8		8
9	1 FILE CREATED	9
10		10
11	190 SYMBOLS	11
12		12
13		13
14		14
15		15
16		16
17		17
18		18
19		19
20		20
21		21
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1		1
2		2
3		3
4		4
5	00000	5
6	00100	6
7		7
8	00200	8
9	00300	9
10		10
11	00400	11
12	00500	12
13		13
14	00600	14
15	00700	15
16		16
17	01000	17
18	01100	18
19		19
20	01200	20
21	01300	21
22		22
23	01400	23
24	01500 00000000 00000000 00000000 00000000 00000011 11111111 11111111 11111111	24
25		25
26	01600	26
27	01700	27
28		28
29	02000	29
30	02100	30
31		31
32	02200	32
33	02300	33
34		34
35	02400	35
36	02500	36
37		37
38	02600	38
39	02700	39
40		40
41	03000	41
42	03100	42
43		43
44	03200	44
45	03300	45
46		46
47	03400	47
48	03500	48
49		49
50	03600	50
51	03700	51
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1		1
2		2
3		3
4		4
5	04000	5
6	04100	6
7		7
8	04200	8
9	04300	9
10		10
11	04400	11
12	04500	12
13		13
14	04600	14
15	04700	15
16		16
17	05000	17
18	05100	18
19		19
20	05200	20
21	05300	21
22		22
23	05400	23
24	05500	24
25		25
26	05600	26
27	05700	27
28		28
29	06000	29
30	06100	30
31		31
32	06200	32
33	06300	33
34		34
35	06400	35
36	06500	36
37		37
38	06600	38
39	06700 00000000 00000000 00000000 00000000 00000000 00000000 10000000 00000000	39
40		40
41	07000	41
42	07100	42
43		43
44	07200	44
45	07300	45
46		46
47	07400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	47
48	07500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	48
49		49
50	07600	50
51	07700	51
52		52
53		53
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1		1
2		2
3		3
4		4
5	20000	5
6	20100	6
7		7
8	20200	8
9	20300	9
10		10
11	20400	11
12	20500	12
13		13
14	20600	14
15	20700	15
16		16
17	21000	17
18	21100	18
19		19
20	21200	20
21	21300	21
22		22
23	21400	23
24	21500	24
25		25
26	21600	26
27	21700	27
28		28
29	22000	29
30	22100	30
31		31
32	22200	32
33	22300	33
34		34
35	22400	35
36	22500	36
37		37
38	22600	38
39	22700	39
40		40
41	23000	41
42	23100	42
43		43
44	23200	44
45	23300	45
46		46
47	23400	47
48	23500	48
49		49
50	23600	50
51	23700	51
52		52
53		53
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1		1
2		2
3		3
4		4
5	24000 11111111 11111111 11111111 11111111 11100000 00000000 00000000 00000000	5
6	24100	6
7		7
8	24200	8
9	24300	9
10		10
11	24400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	11
12	24500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111	12
13		13
14	24600	14
15	24700	15
16		16
17	25000	17
18	25100	18
19		19
20	25200	20
21	25300	21
22		22
23	25400	23
24	25500	24
25		25
26	25600	26
27	25700	27
28		28
29	26000	29
30	26100	30
31		31
32	26200	32
33	26300	33
34		34
35	26400	35
36	26500	36
37		37
38	26600	38
39	26700	39
40		40
41	27000	41
42	27100	42
43		43
44	27200	44
45	27300	45
46		46
47	27400	47
48	27500	48
49		49
50	27600	50
51	27700	51
52		52
53		53
54		54
55		55
56		56
57		57
58		58